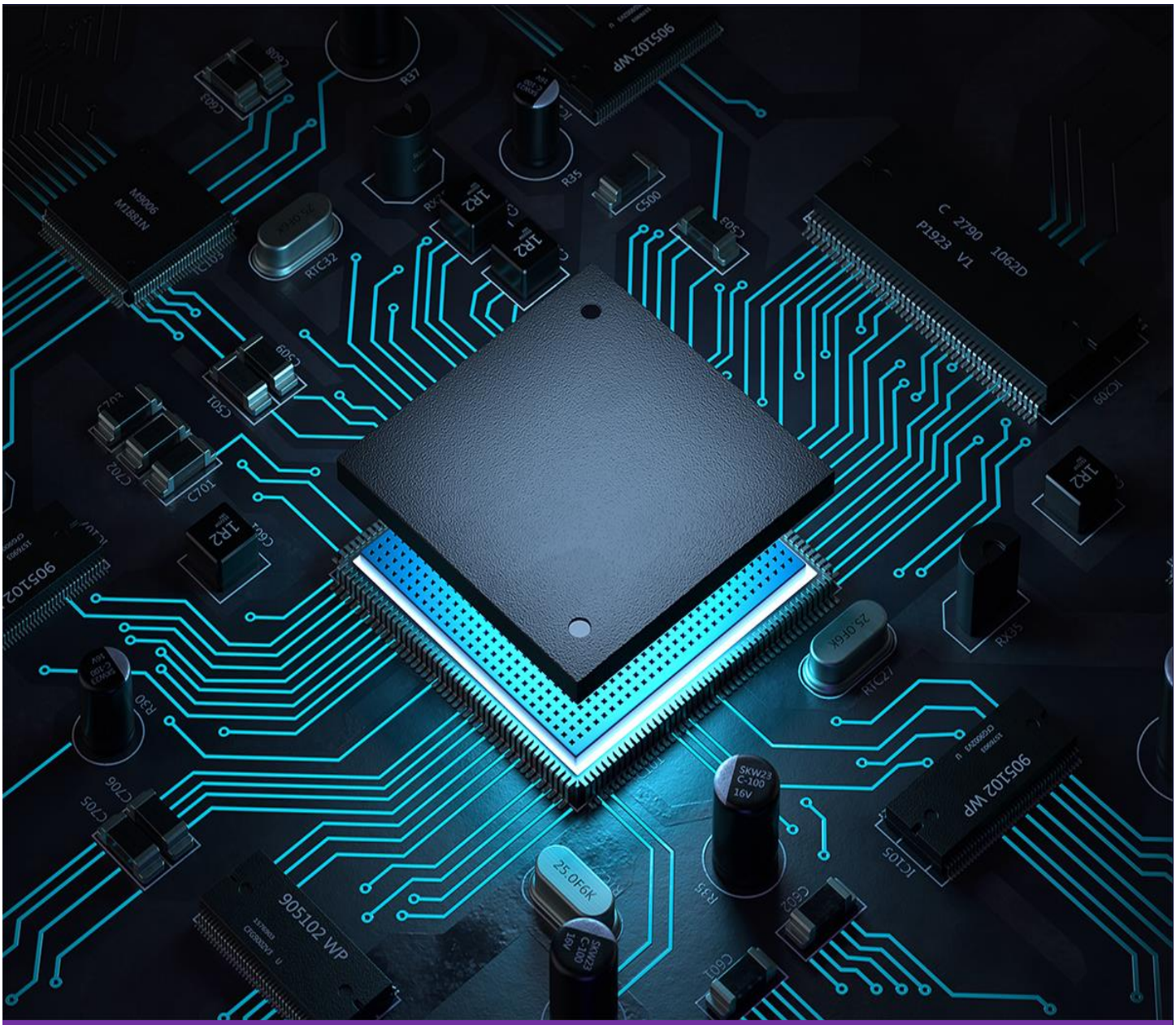


مقدم لكم من فريق سبكتروم

ملخص مختبرPLC

عمل الطالب عبدالله شاويش

بناءا على مبدأنا الراسخ في خدمة الطلاب نقدم لكم ملخص مختبر PLC بأسلوب وشرح استثنائي - نؤكد ان هذا العمل لا يغني عن المرجع الرسمي للكلية – وان هذا عمل شخصي قد ان لا يخلو من الاخطاء

المقدمة

تعريف عام

PLC اختصار (programmable logic controller) وتعني المتحكم المنطقي المبرمج ويمكن تعريف الـ PLC أنه جهاز إلكتروني رقمي مع ذاكرة قابلة للبرمجة لحفظ التعليمات الخاصة لتنفيذ وظائف ومهام مثل العمليات المنطقية، والتتابع، والتوقيت، والعد، والعمليات الحسابية وغيرها. وذلك للتحكم بالآلات والعمليات الصناعية.

يعتمد الجهاز في عمله على وجود برنامج تحكم يتم حفظه وفق تسلسل معين لتنفيذ خطوات التحكم المطلوب. يقوم الجهاز باستدعاء هذه الخطوات وتنفيذها ومن ثم التحكم بالآلة أو العملية الصناعية بواسطة إشارات الخرج التي يتم إخراجها على أطراف مخارج الجهاز .

أهم الأجزاء الرئيسية لـ PLC هي وحدة التغذية بالطاقة، وحدات الدخل والخرج من أجل إدخال المعلومات من أجهزة الدخل أو إعطاء الأوامر لتشغيل أجهزة الخرج. وحدة المعالجة المركزية CPU وتتكون من عنصرين أساسيين وهما المعالج والذاكرة ويختلف التركيب الدقيق لوحدة المعالجة المركزية من صانع إلى آخر. ويبين الشكل أدناه تركيب جهاز (PLC) ويشبه التركيب الداخلي لجهاز الكمبيوتر.

1 -مصدر التغذية :power supply تقوم هذه الوحدة بتوفير الجهود المناسبة لعمل المتحكم المنطقي المبرمج(PLC) ، حيث توفر الجهود الملائمة لعمل وحدة الـ CPU ووحدات الدخل والخرج والحساسات وغيرها. وتشمل هذه الوحدة جميع المراحل اللازمة مثل تحويل الجهد الداخل المتردد إلى مستمر أو التوحيد أو الترشيح والتنظيم.

2-وحدة الدخل :Input Module يتم توصيل وحدة الدخل بمجموعة من العناصر الفيزيائية مثل المفاتيح الكهربائية والحساسات ومقاييس الحرارة والوزن وحساسات مستوى السوائل وغيرها، حيث تقوم وحدة الدخل باستقبال الإشارات التماثلية والرقمية المرسلة من هذه العناصر وتقوم بتحويلها إلى إشارات منطقية يمكن ان تتعامل معها وحدة المعالجة المركزية. وتنقسم إلى نوعين:

وحدات الإدخال الرقمية: وهذه الوحدات مخصصة لاستقبال الإشارات الرقميةON ، OFF(0 أو 1)، القادمة من عناصر الدخل التي لها حالتا تشغيل فقط، مثل الضواغط، والمفاتيح الحدية، والمفاتيح التقاربية(Sensors proxirnity) ، والخلايا الضوئية، والعوامات الكهربائية. وتقوم هذه الوحدات بنقل المعلومات عن عناصر الدخل إلى وحدة المعالجة المركزية. وتعمل وحدات الدخل بجهود مستمرة (dc) أو متغيرة (ac) وبقيم مختلفة ومتنوعة مثل 11 OV،24V.24OV

وحدات الإدخال التماثلية: وتقوم باستقبال الإشارات التماثلية التي تتغير ضمن مجال معين والواردة من الحساسات المختلفة. وهذه الإشارات لا تتخذ قيمتين فقط، بل يمكن أن تأخذ عددًا لا نهائي من القيم ضمن مجال خرج هذه الحساسات. ويختلف خرج المجسات المختلفة من ناحية النوع والقيمة. فبعضها يعطي فرق جهد قليل بالميلي فولت مثل الازدواجات الحرارية، أو عدة فولتات مثل المحولات الخطية. وبعضها تتغير فيه المقاومة الكهربائية بشكل واسع مثل الثرمستور أو بقيم صغيرة. ومجسات أخرى تعطي تيارًا متغيرًا على مخرجها. ولذلك تقوم الشركات الصانعة بتصميم وحدات إدخال متنوعة تختص بإدخال أنواع الإشارات المختلفة وضمن مجال قياسي محدد.

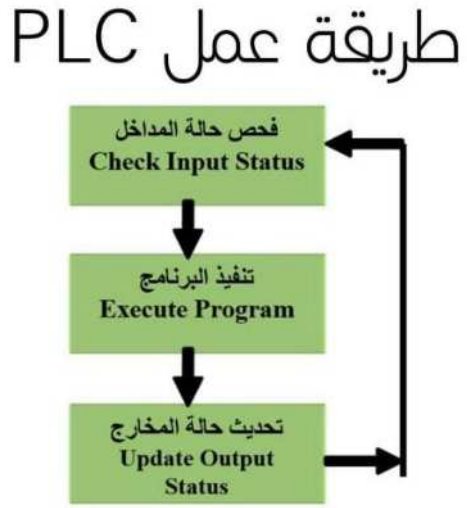
3-وحدات الإخراج :output Module وهي وحدات تقوم بإخراج الإشارات الكهربائية المطلوب إخراجها من وحدة المعالجة المركزية، وذلك لتسيير العملية الصناعية حسب البرنامج المدخل للجهاز من المستخدم. وتعمل على تحويل الإشارة الخارجة من وحدة المعالجة إلى إشارة ذات جهد مناسب للتحكم بعمل المشغلات (رقمية، تماثلية). وهناك عدة أنواع من وحدات الخرج منها:

وحدة الخرج الرقمية: وهي وحدة الخرج التي تأخذ إشارة الخرج فيها إحدى حالتين (1 أو 0) و عادة توجد وحدات الخرج الرقمية بإحدى الأنواع التالية: 1) وحدة الخرج بترياك وتعمل مع الجهود المتغيرة. 2) وحدة الخرج بترانزستور ويكون الترانزستور إما من النوع NPN أو PNP وتستخدم مع الجهود المستمرة، ويمكن تشغيلها على سرعات عالية. 3) وحدة الخرج بمرحل: ويمكن تشغيلها على الجهود المستمرة أو المتغيرة. وتوفر عزلًا بين دارة تشغيل المرحل والحمل، وتقاوم التغيرات المفاجئة للتيارات والحالات العابرة للجهود. ولكن المرحل بطيء العمل ذو حجم كبير نسبيًا، ويتلف أسرع من المفاتيح الساكنة الأخرى كالترانزستور والترياك.

وحدات الخرج التماثلية: تكون الإشارة الخارجة من وحدة الخرج إشارة تماثلية تتغير ضمن مجال معين من التيارات أو الجهود (0-10V) أو(0-20mA) ، وتستخدم هذه الوحدة للتحكم بالصمامات والمشغلات التماثلية ونظم قيادة محركات التيار المتناوب والمستمر.(DC & AC Drives)

4-المعالج أو وحدة التحكم المركزية :CPU هو "عقل" الـ PLC. عادة ما يتكون الـ CPU من معالج دقيق لتنفيذ الأوامر المنطقية والتحكم في الاتصالات بين الوحدات. يتطلب المعالج ذاكرة لتخزين تعليمات برنامج المستخدم والقيم العددية وحالة أجهزة الإدخال/الإخراج. يتحكم المعالج في جميع أنشطة الـ PLC وهي مصممة بحيث يمكن للمستخدم إدخال البرنامج المطلوب. ويتم تنفيذ برنامج PLC كجزء من عملية متكررة يشار إليها بالـ scan. وتختلف المعالجات بطول الكلمة (word length) التي تعالجها بالبت (32، 16، 8، 4) وبسرعة تردد الساعة (Clock) التي تعمل عليها بالميغا هيرتز.

5- برنامج التحكم أو الـ (Software): هي مجموعة من الأوامر المطلوب تنفيذها بالتتابع المكتوبة به لتنفيذ عملية التحكم، ويخزن البرنامج على وحدات التخزين مثل الأقراص المرنة والصلبة. وعند الرغبة في التعديل في برنامج التحكم يتم التعديل في البرنامج دون الحاجة إلى تغيير نظام التحكم. باستخدام لغة برمجة مثل ladder وهي أحد أكثر لغات برمجة PLC شيوعًا.



كيف يعمل PLC

تتضمن الأجزاء الأساسية لـ PLC: وحدات الإدخال والإخراج، ووحدة معالجة مركزية (CPU)، وجهاز برمجة. تتكون وحدة الإدخال من الأجهزة التي تراقب الآلات أو المستخدمة في التحكم في العملية. قد تكون هذه الأجهزة رقمية أو تناظرية، مثل المجسات (Sensors)، مفاتيح الحد(limit switches) ، مفاتيح الإيقاف والتشغيل، وبادئ تشغيل المحرك (motor starters) وغيرها.

تكون قيم الدخل الرقمية إما في وضع التشغيل أو الإيقاف، وتمثل الإشارات التناظرية ظروف الماكينة أو العملية كمجموعة من قيم الجهد أو التيار . تتمثل الوظيفة الأساسية لدائرة إدخال PLC في تحويل الإشارات التي توفرها هذه المفاتيح وأجهزة الاستشعار المتنوعة إلى إشارات منطقية يمكن أن تستخدمها وحدة المعالجة المركزية،

تقوم وحدة المعالجة المركزية (CPU) بتقييم حالة المدخلات والمخرجات والمتغيرات الأخرى أثناء قيامها بتنفيذ البرنامج المخزن مسبقًا فيها. ثم ترسل وحدة المعالجة المركزية إشارات لتحديث حالة المخرجات. وتقوم وحدات الإخراج بتحويل إشارات التحكم من وحدة المعالجة المركزية إلى قيم رقمية أو تناظرية يمكن استخدامها للتحكم في أجهزة الإخراج المختلفة مثل المحرك الكهربائي.

تُعرف عملية القراءة المتتابعة للمدخلات وتنفيذ البرنامج في الذاكرة وتحديث المخرجات باسم المسح scanning. يوضح الشكل في الأعلى تمثيل رسومي لعملية الـ scan.

مميزات PLC

1-تكلفة أقل وحجم أصغر، وسرعة في الأداء.

2-سهولة صيانتها ومعرفة الخطأ إن وجد.

3-توافر عدد كبير من المرحلات الداخلية والمؤقتات والعدادات وغيرها من الوظائف في جهاز واحد.

4-عدد كبير من الملامسات المساعدة حيث يمكن تكرار استخدام نفس العنوان لعدد كبير من المرات.

5-سهولة تنفيذ التعديلات وإصلاح الأخطاء: في لوحات كلاسيك كنترول التقليدية فإن أي تغيير في عمل نظام التحكم يستلزم تغيير التوصيلات مما يتطلب وقتًا وجهدًا، بينما يتم ذلك بواسطة تعديل البرنامج فقط في جهاز الـPLC.

6-سهولة متابعة العملية الصناعية خلال العمل من خلال البرنامج على جهاز الحاسوب وتوثيقها.

7-الاعتمادية والموثوقية: حيث إن لوحات كلاسيك كنترول التقليدية تعتمد في عملها على المرحلات ذات طبيعة العمل الميكانيكية.

استخدامات PLC

منذ اختراعها تم استخدام PLC في كل قطاع تقريبًا من قطاعات الصناعة، بما في ذلك مصانع الصلب ومصانع الورق ومصانع الأغذية والمصانع الكيماوية ومحطات الطاقة الكهربائية. تقوم PLCs بأداء مجموعة كبيرة ومتنوعة من مهام التحكم، من التحكم المتكرر في التشغيل/الإيقاف للآلات البسيطة إلى التصنيع المتطور والتحكم في العمليات المعقدة.

بسواعدنا نبنيها

PLC Symbols

NO Relay Contact



NC Relay Contact



Output Coil



One Shot - Positive Edge Detection



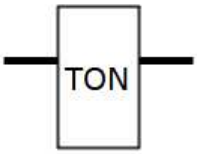
Set Coil



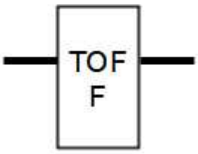
Reset Coil



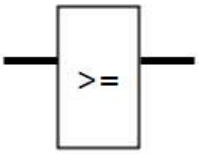
Timer Delay ON



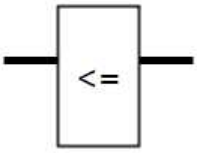
Timer Delay OFF



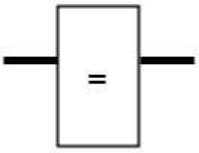
Greater Than or Equal to



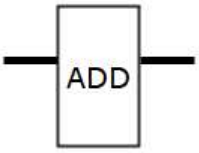
Less Than or Equal to



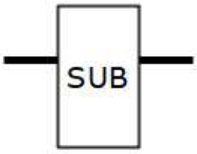
Equal to



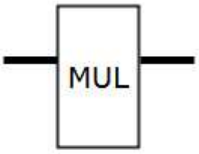
Addition



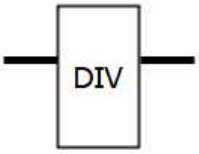
Subtractor



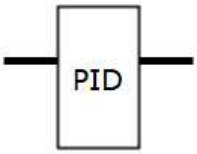
Multiplication



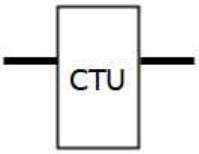
Division



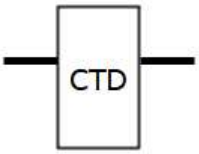
PID Controller

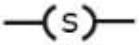
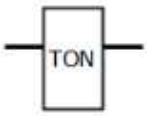


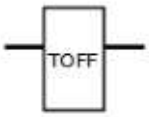
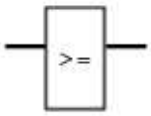
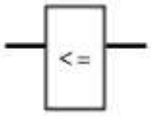
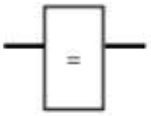
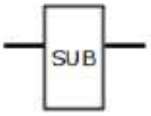
Counter Up

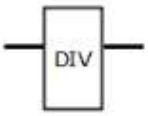
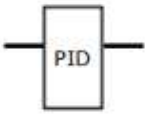
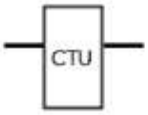
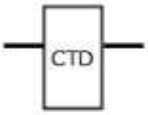


Counter Down



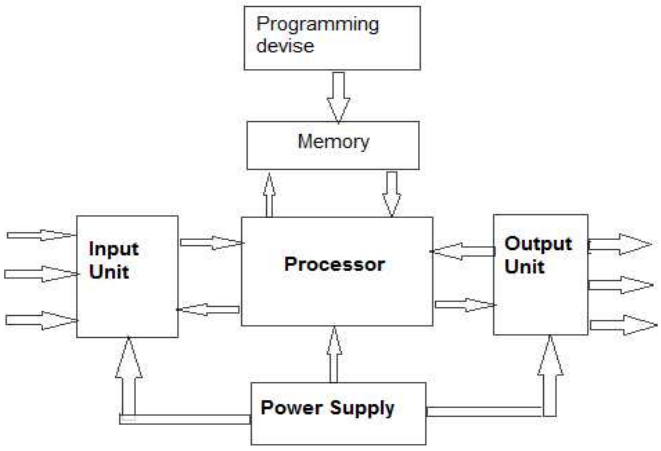
Symbol	Name	Description
	Normally Open Contact (NO) Symbol	Its operation is very simple. When the condition is true then the contact is closed and output logic flow is enabled. When the condition is FALSE then the contact is OPEN and output logic flow is blocked.
	Normally Closed Contact (NC) Symbol	When the condition is true then the contact is OPEN and output logic flow is blocked. And when the condition is FALSE then the contact is CLOSED and output logic flow is enabled. The NC contact symbol operation is nothing but just the opposite to the NO contact symbol.
	Output Coil Symbol	When the input condition is true then the output is ON. And when the input condition is false then the output is OFF.
	One Shot Symbol– Positive Edge Detection	When the input condition transitions from FALSE to TRUE then the output is ON, for the time taken to do one PLC scan.
	Set Coil Symbol	When the input condition transitions from false to true then the output is set ON and once the output is set ON, it stays ON, even if the input condition goes FALSE.
	Reset Coil Symbol	When the input condition is true then the output is reset to OFF. And when the input condition is false it has no effect on the output. The SET and RESET coils can share the same variable address and therefore work hand in hand.
	Timer Delay on Symbol	When the input condition is true then the timer begins. And when the present time set point has been reached the output turns ON. If the input condition goes FALSE, at any stage, the timer stops and the output turns OFF as well.

	Timer Delay off Symbol	When the input condition is true then the output turns ON. Then if the input condition goes false the timer begins and when the preset time set point has been reached the output turns OFF. If the input condition goes TRUE, at any stage, the timer stops and the output turns ON as well.
	Greater than or Equal to	When the two inputs comparison is true then output is true otherwise false.
	Less than or Equal to	When the two inputs comparison is true then output is true otherwise false.
	Equal to	When two inputs are equal then output are true.
	Adder	Adder is used to add two or more inputs.
	Subtractor	Subtractor is used to subtract one value from the other one.
	Multiplication	Multiplication block is used to multiply two or more values.

	Division	Division block is used to divide two values.
	PID Controller	PID Controller is used to control the process value. The process variable is measured from the primary element (Input) and the output is manipulated to maintain the process variable value at the set point input value. The Proportional, Integral and Derivative input values are tune to increase the performance of controlling. Which is done according to process requirements.
	Counter Up	When input is True from False then counter increases 1 value from the last stored value. When this stored value reaches the preset value, the output will go True and the stored value is set again to 0.
	Counter Down	When input is True from False then counter decreases 1 value from the last stored value. When this stored value reaches 0 value, the output will go True and the stored value is set again to preset value.

التجربة الاولى

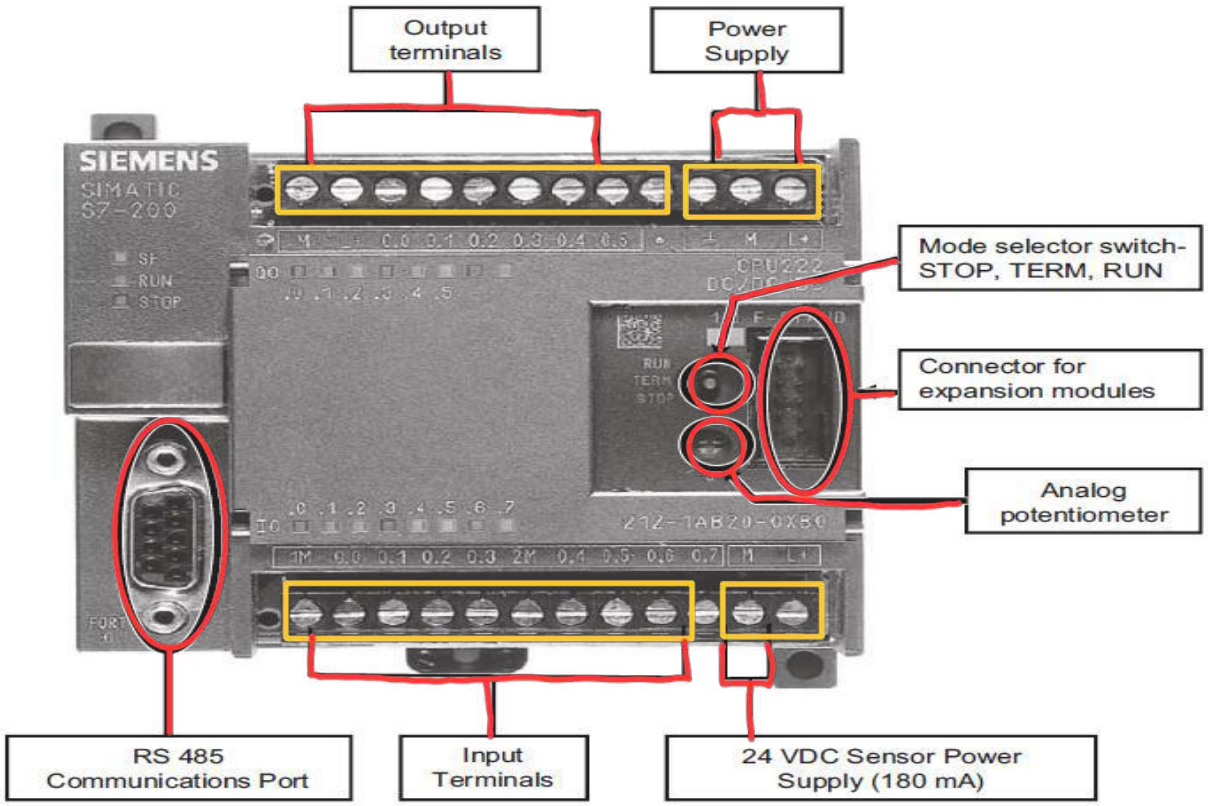
MICRO/WIN and S7-200 Simulation



The PLC System

PLC Programming Language

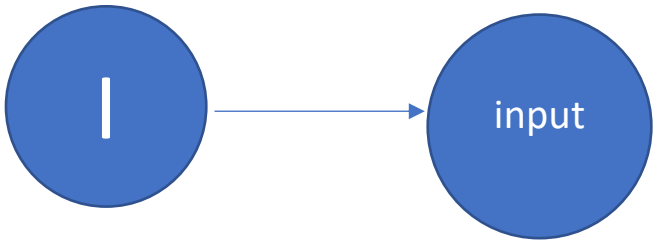
- Ladder diagram(LAD)
- Statement List (STL)
- Functional Block(FB)

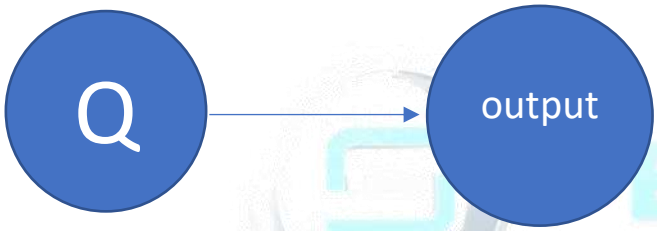


Inputs will be defined as [I0.0; I0.1; I0.2; I0.3; …..]

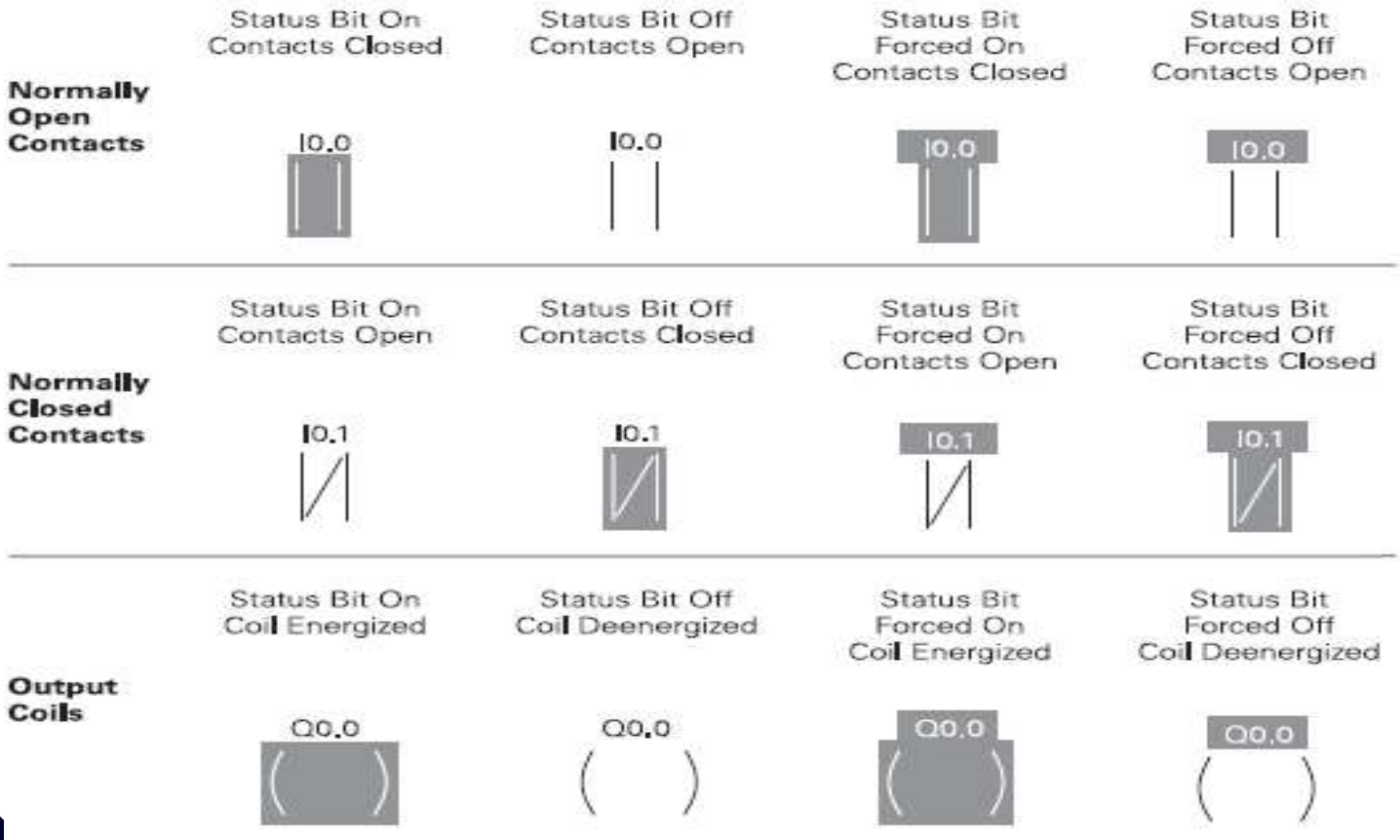
Outputs will be defined as [Q0.0; Q0.1; Q0.2; Q0.3; …….]

Flags will be defined as [M0.0; M0.1; M0.2; M0.3; …….]

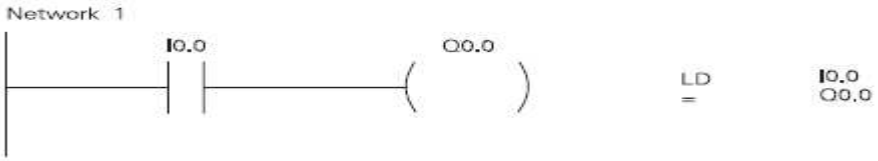




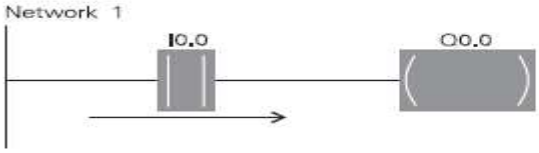
MECHATRONICS SPECTRUM



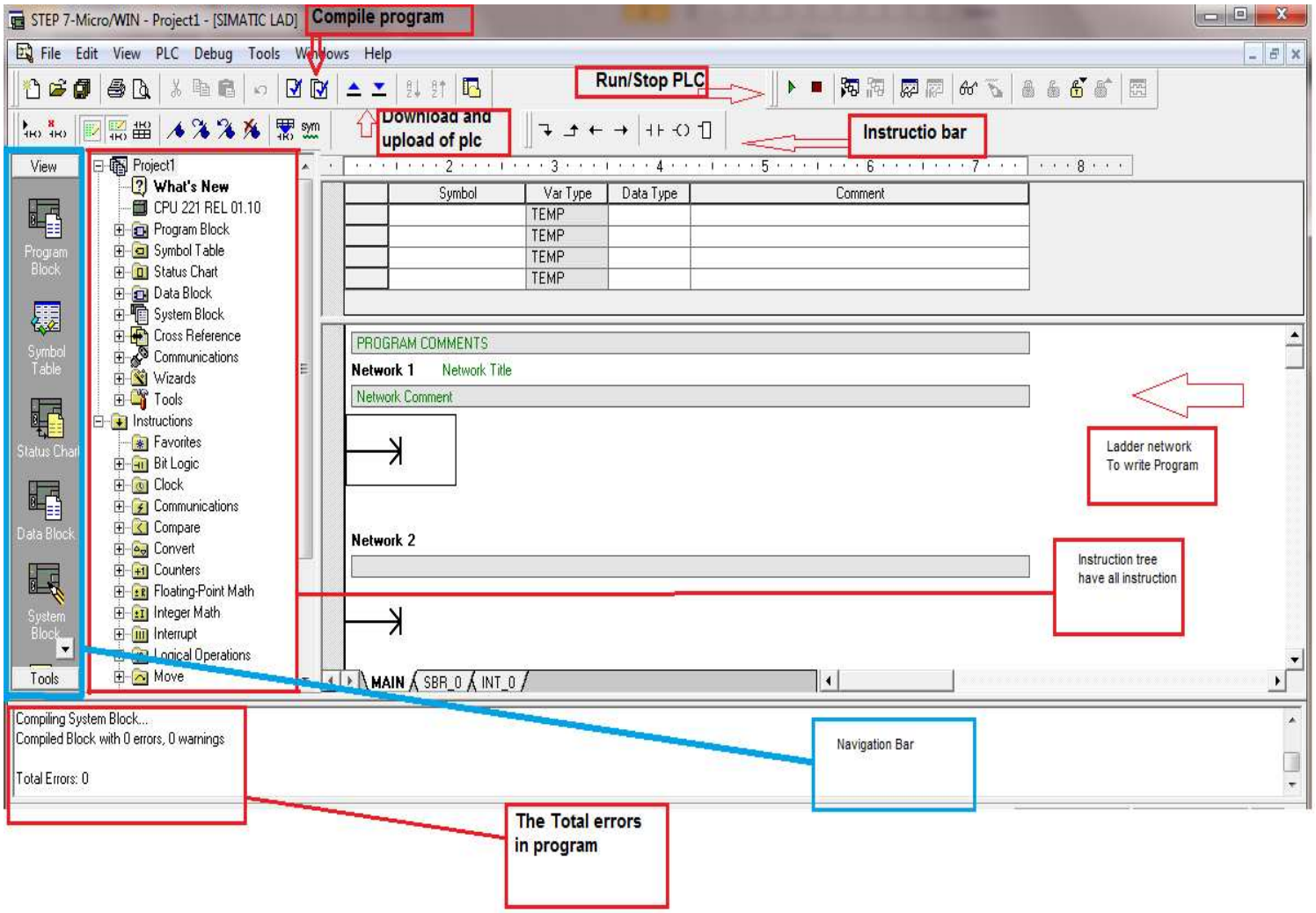
When the switch is open the CPU receives a logic 0 from input I0.0. The CPU sends a logic 0 to output Q0.0 and the light is off.

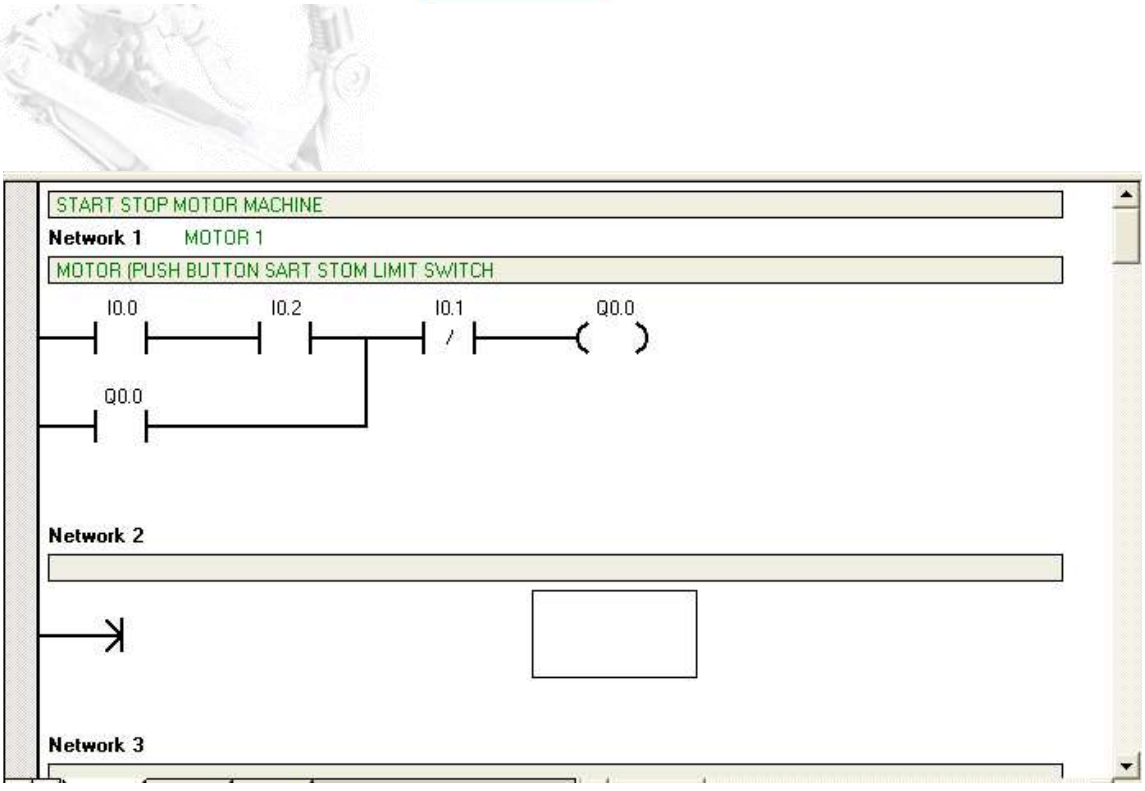


When the switch is closed the CPU receives a logic 1 from input I0.0. The CPU sends a logic 1 to output Q0.0, thus activating Q0.0. The light turns on.



R O N I C S
RUM
T E A M


STEP 7-Micro/WIN - Project1 - [SIMATIC LAD]
Compile program
File Edit View PLC Debug Tools Windows Help
Run/Stop PLC
Download and upload of plc
Instructio bar
View
Project1
What's New
CPU 221 REL 01.10
Program Block
Symbol Table
Status Chart
Data Block
System Block
Cross Reference
Communications
Wizards
Tools
Instructions
Favorites
Bit Logic
Clock
Communications
Compare
Convert
Counters
Floating-Point Math
Integer Math
Interrupt
Logical Operations
Move
Symbol
Var Type
Data Type
Comment
TEMP
PROGRAM COMMENTS
Network 1
Network Title
Network Comment
Network 2
Ladder network To write Program
Instruction tree have all instruction
MAIN
SBR_0
INT_0
Compiling System Block...
Compiled Block with 0 errors, 0 warnings
Total Errors: 0
Navigation Bar
The Total errors in program


START STOP MOTOR MACHINE
Network 1
MOTOR 1
MOTOR (PUSH BUTTON SART STOM LIMIT SWITCH
I0.0
I0.2
I0.1
Q0.0
Q0.0
Network 2
Network 3

The steps thinking in reviews of the PLC program

1. The understanding and analysis of the industrial process to be the work of the program

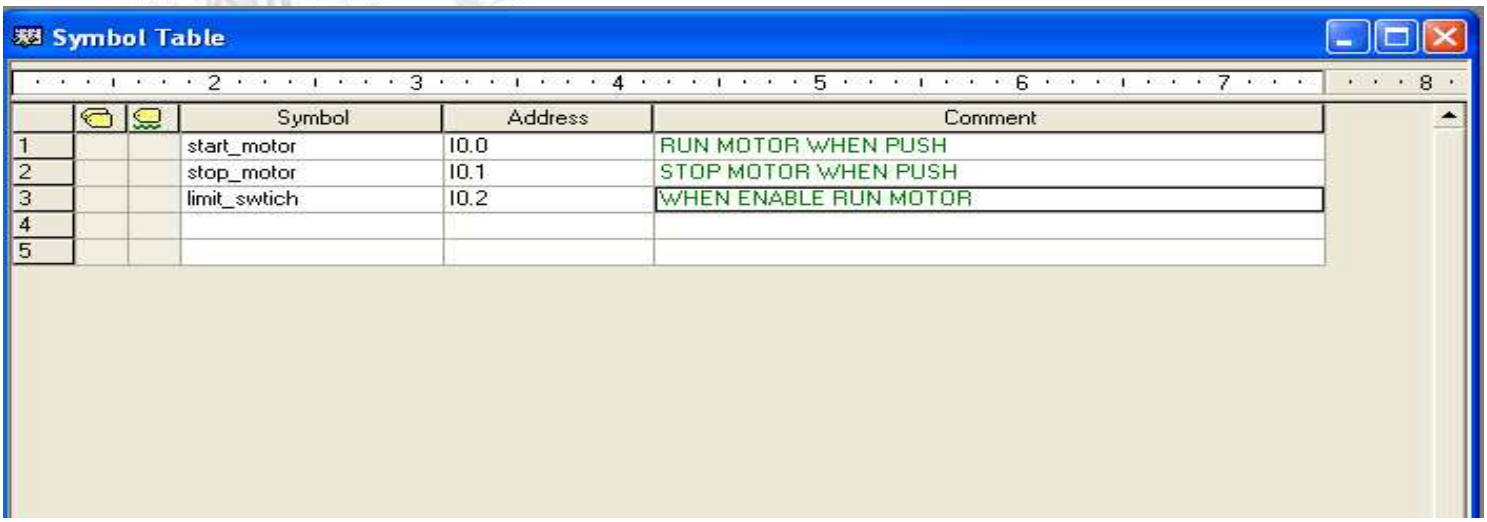
2. The identification of the Input and Output in the table with the description

3. Division the industrial process to the parts and work program of each part

MECHATRONICS SPECTRUM TEAM

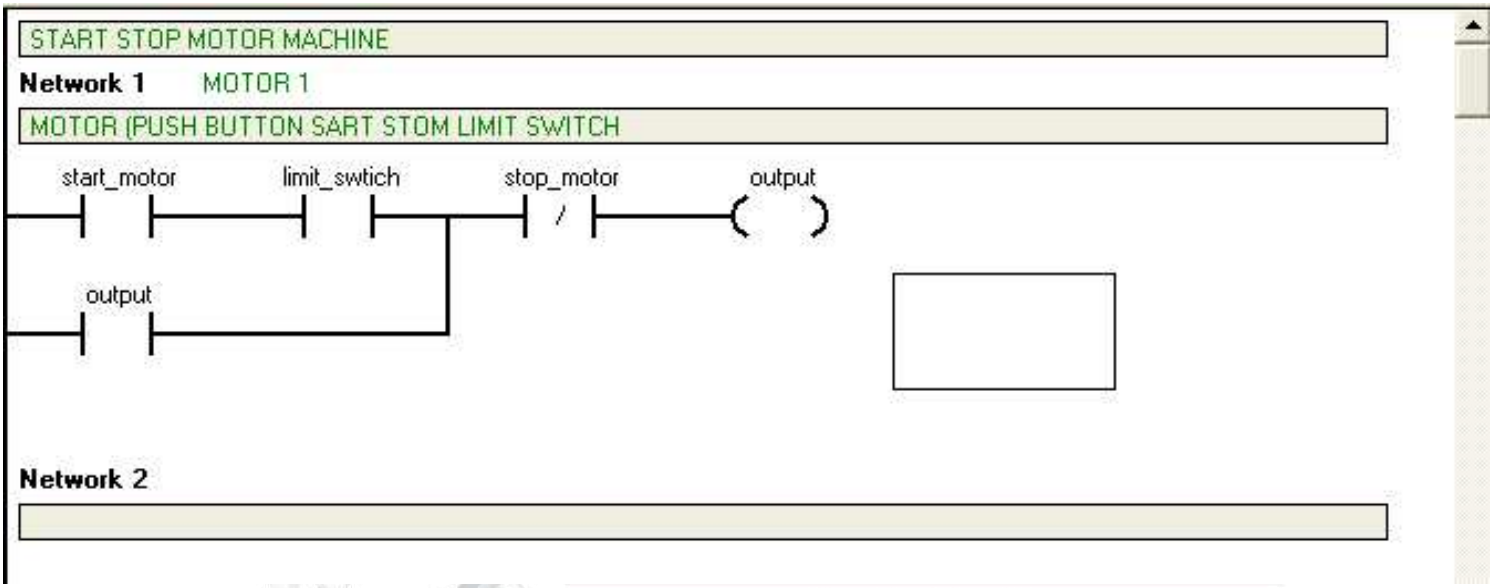
Write the symbol tables (it easy to search process within the program)

View → component→ symbol table



Now, if we want appearance of detail the program of data entered by the symbol table is work the following:

View → symbol table → apply symbol to project



If we are to reverse previous situation

view→ symbolic addressing

After we write the program, we want to compile the program for errors

From PLC → compile all

We note in the Output window, [total error: 0]

To save program

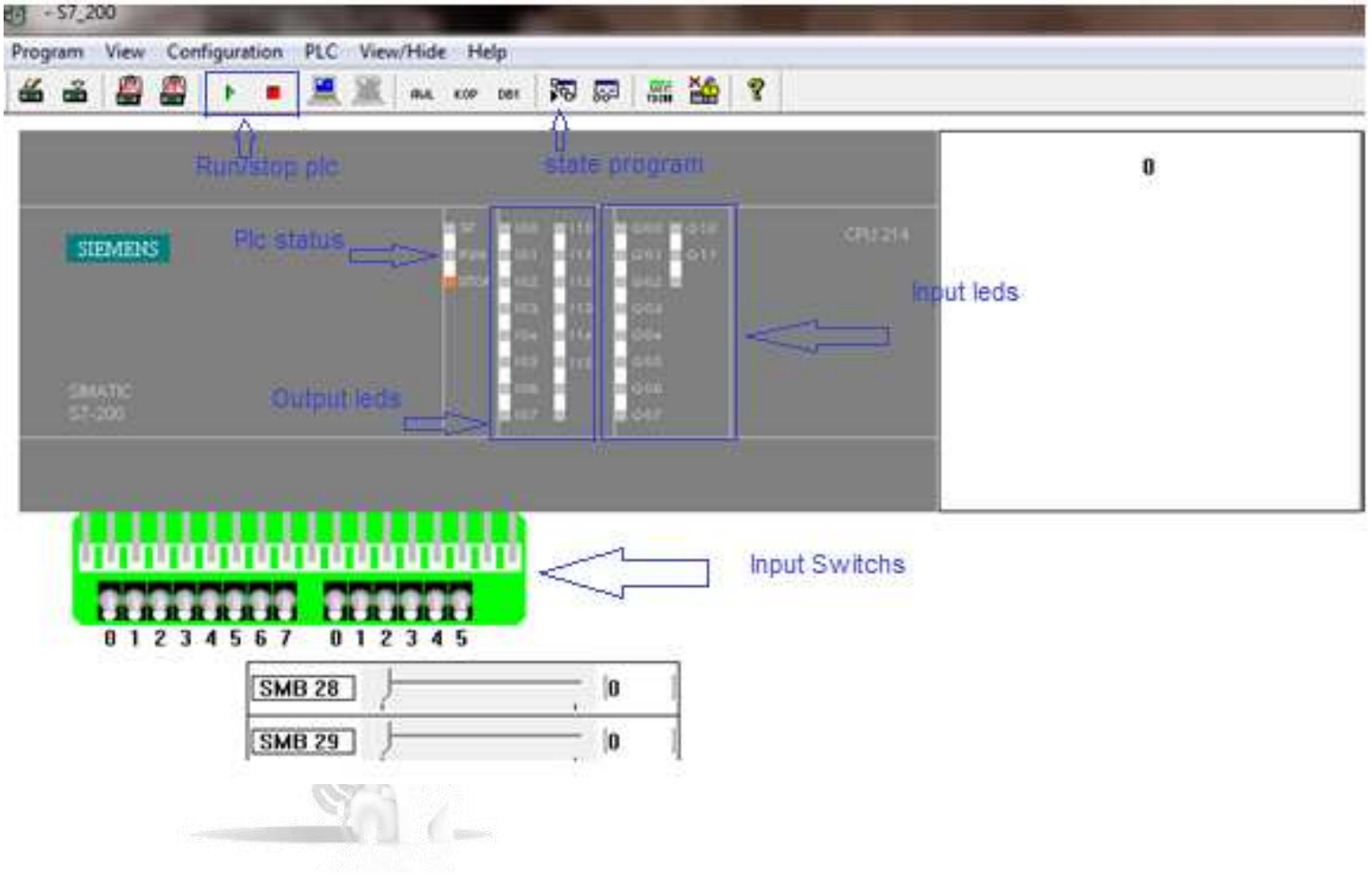
File→ Export [To save as (awl) file]

S7-200 Simulation

To load the program to S7-200 simulator

Program → Load Program

Tow windows are open one the program in ladder diagram and the second in Instruction list.



To Run / stop simulation of PLC from PLC we press Run or stop.

رابط تحميل البرنامج :

https://plc247.com/download-step7-microwin-v4sp9-for-plc-s7-200-siemens-googledrive

التجربة الثانية

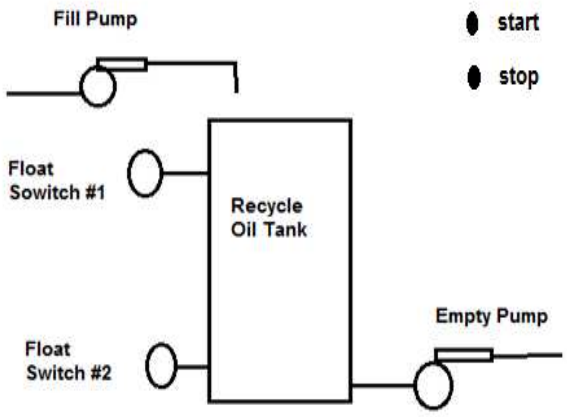
MICRO/WIN and S7-200 Simulation Problems

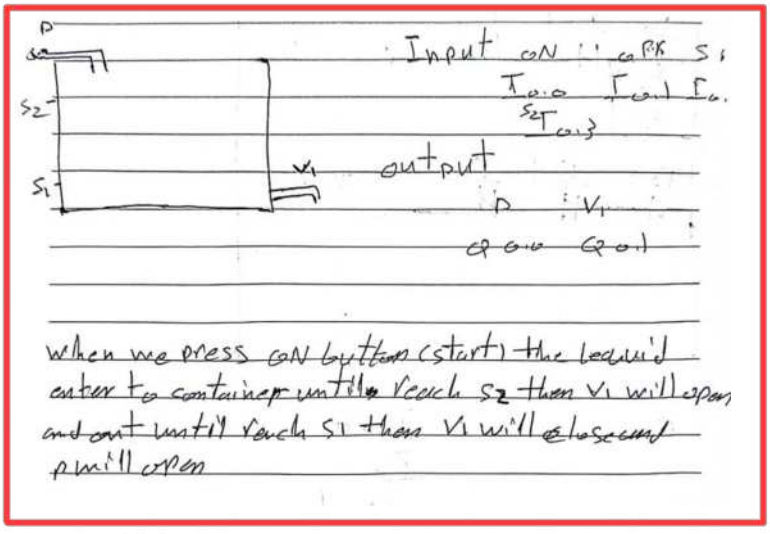
Problem 1

In a recycle oil tank problem shown in the figure below, when the "Fill" button is pressed, the Fill Pump should run until Float Switch #1 is activated or the stop button is pressed. When the "Empty "button is pressed and Float Switch #2 is activated, the Empty Pump should run until Float Switch #2 is de-activated or the Stop button is pressed.

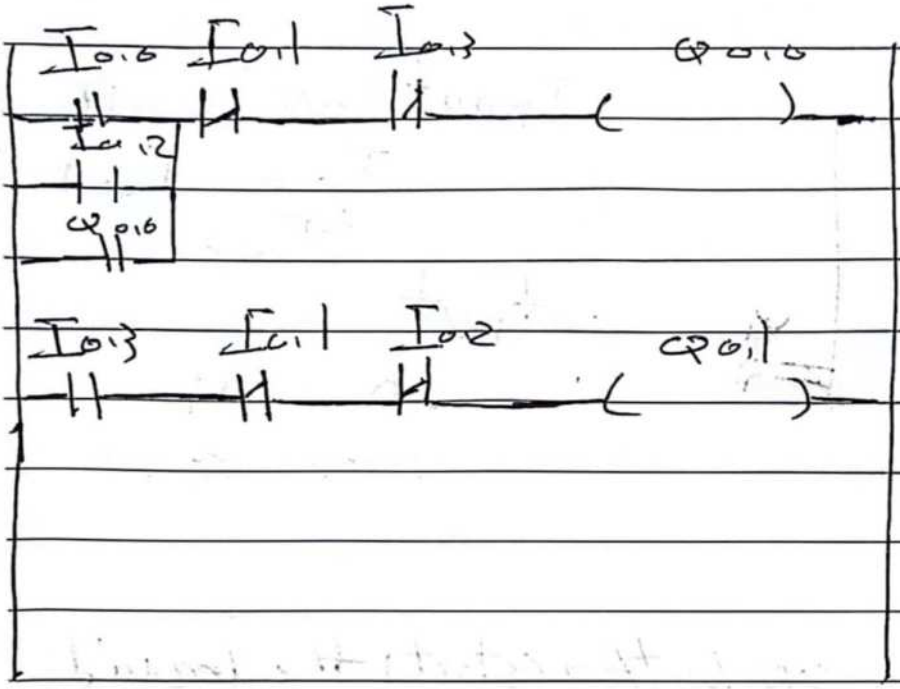
-Write the ladder diagram and input/output table.

-Write the program in Instruction list.





I/O list



Ladder digram.

LD	I0.0
O	I0.2
O	Q0.1
AN	I0.1
AN	I0.3
=	Q0.6
LD	I0.3
O	Q0.1
AN	I0.1
AN	I0.2
=	Q0.1

instruction list.

بسواعدنا نبنيها
MECHATRONICS SPECTRUM TEAM

Problem 2

In the figure below, when we pressed start button the Fill Pump P1,P2,P3 are run.

When S2 is activated or the stop button pressed P1 Stop,

When S3 is activated or the stop button pressed P2 Stop ,

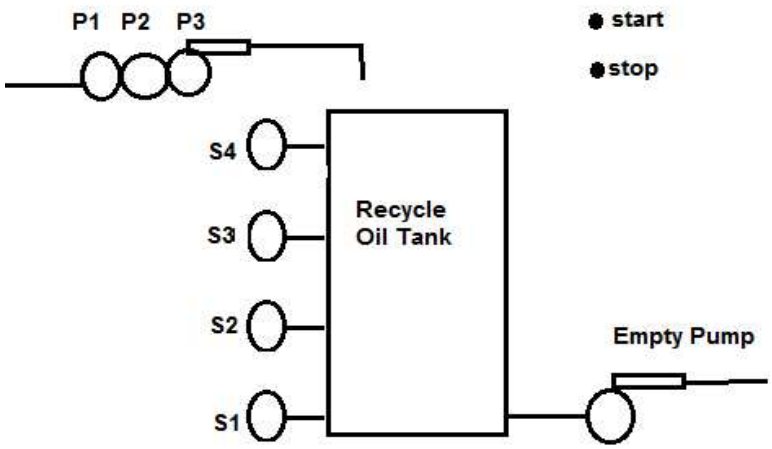
When S4 is activated or the stop button pressed P2 Stop ,

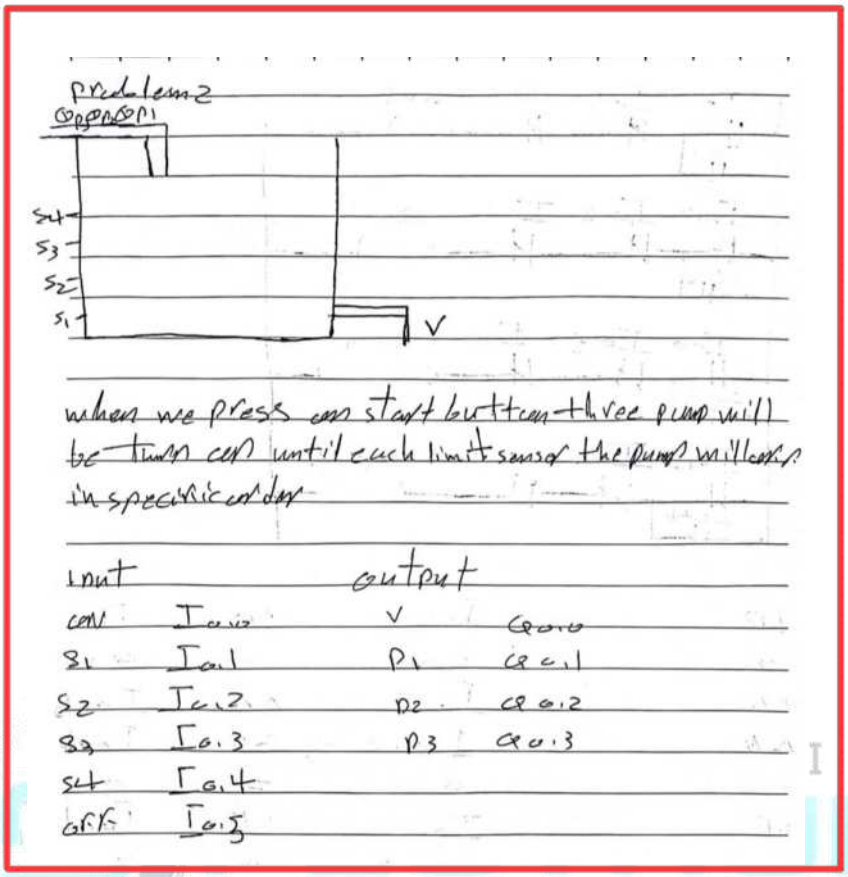
When S4 is activated the Empty Pump should run until S1 de-activated or stop button pressed.

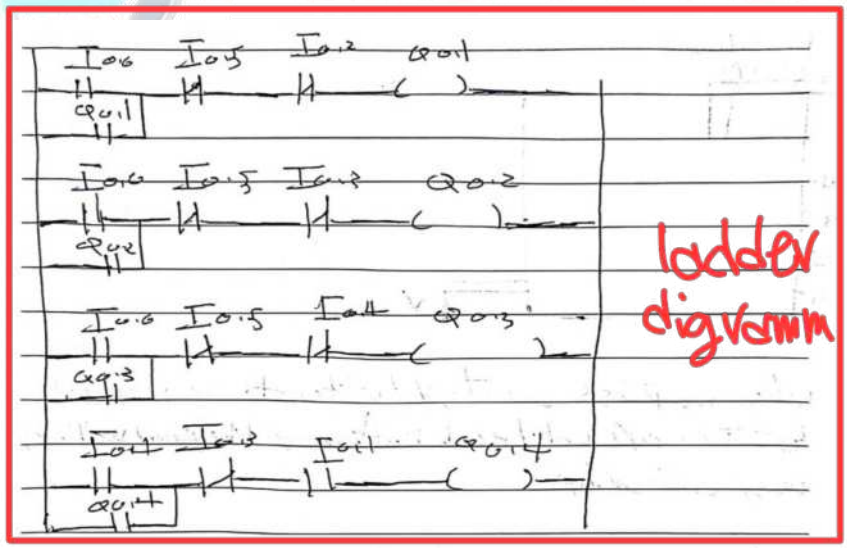
MECHATRONICS SPECTRUM TEAM

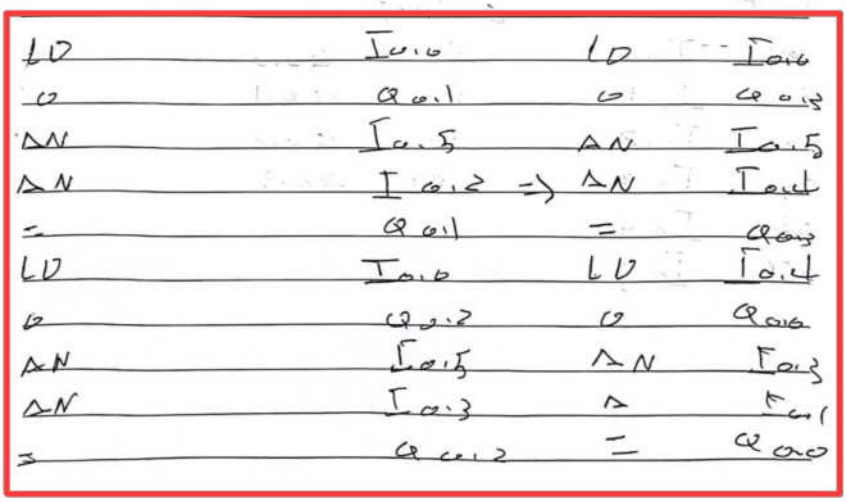
-Write the ladder diagram and input/output table.

-Write the program in Instruction list.




problem2
S4
S3
S2
S1
V
when we press on start button three pump will be turn on until each limit sensor the pump will work in specific order
Input
output
con I0.0 V Q0.0
S1 I0.1 P1 Q0.1
S2 I0.2 P2 Q0.2
S3 I0.3 P3 Q0.3
S4 I0.4
off I0.5


I0.6 I0.5 I0.2 Q0.1
Q0.1
I0.6 I0.5 I0.3 Q0.2
Q0.2
I0.6 I0.5 I0.4 Q0.3
Q0.3
I0.4 I0.3 I0.1 Q0.4
Q0.4
ladder digramm


LD I0.6 LD I0.6
O Q0.1 O Q0.2
AN I0.5 AN I0.5
AN I0.2 => AN I0.4
= Q0.1 = Q0.3
LD I0.6 LD I0.4
O Q0.2 O Q0.0
AN I0.5 AN I0.3
AN I0.3 A I0.1
= Q0.2 = Q0.0
instruction list

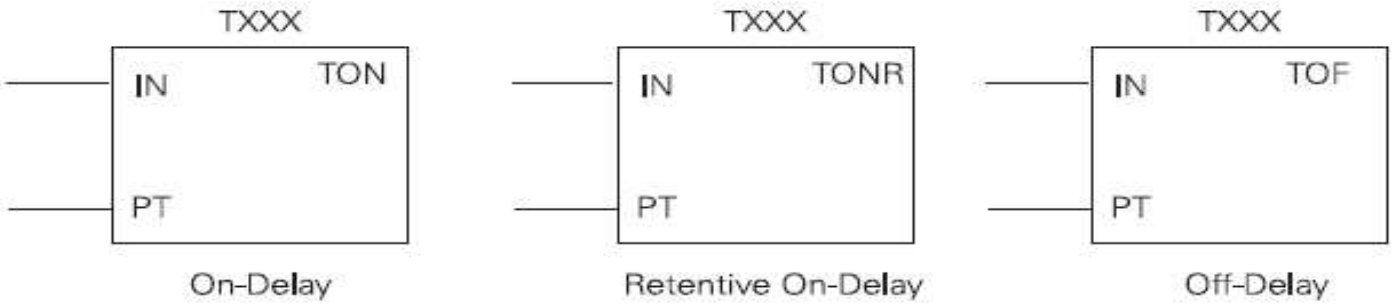
التجربة الثالثة

Timers

Timer Instructions

On-Delay Timer / Retentive On-Delay Timer

The On-Delay Timer (TON) and Retentive On-Delay Timer (TONR) instructions count time when the enabling input is on. The timer number (Txx) determines the resolution of the timer, and the resolution is now shown in the instruction box.

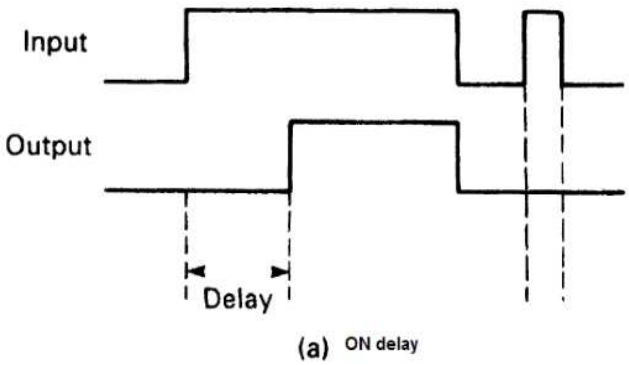


Off-Delay Timer

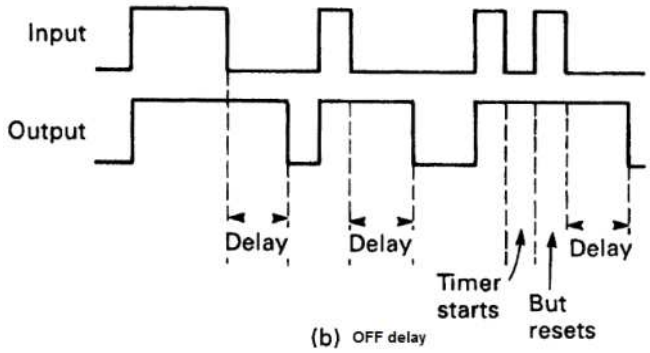
The Off-Delay Timer (TOF) is used to delay turning an output off for a fixed period of time after the input turns off.

The timer number (Txx) determines the resolution of the timer, and the resolution is now shown in the instruction box.

- You can use a TON for timing a single interval.
- You can use a TONR for accumulating a number of timed intervals.
- You can use a TOF for extending time past an off (or false) condition, such as for cooling a motor after it is turned off.



(a) ON delay



(b) OFF delay

Operations of the Timer Instructions

Type	Current >= Preset	State of the Enabling Input (IN)	Power Cycle/First Scan
TON	Timer bit on Current continues counting to 32,767	ON: Current value counts time OFF: Timer bit off, current value = 0	Timer bit off Current value = 0
TONR	Timer bit on Current continues counting to 32,767	ON: Current value counts time OFF: Timer bit and current value maintain last state	Timer bit off Current value can be maintained[1]
TOF	Timer bit off Current = Preset, stops counting	ON: Timer bit on, current value = 0 OFF: Timer counts after on-to-off transition	Timer bit off Current value = 0

The TON and TONR instructions count time when the enabling input is on. When the current value is equal to or greater than the preset time, the timer bit is on.

- The current value of a TON timer is cleared when the enabling input is off, whereas the current value of the TONR timer is maintained when the input is off.

- You can use the TONR timer to accumulate time when the input turns on and off. Use the Reset instruction (R) to clear the current value of the TONR.

- Both the TON and the TONR timers continue counting after the preset is reached, and they stop counting at the maximum value of 32,767 s.

- The TOF instruction is used to delay turning an output off for a fixed period of time after the input turns off. When the enabling input turns on, the timer bit turns on immediately, and the current value is set to 0. When the input turns off, the timer counts until the elapsed time reaches the preset time.

- When the preset is reached, the timer bit turns off and the current value stops incrementing; however, if the input turns on again before the TOF reaches the preset value, the timer bit remains on.

- The enabling input must make an on-to-off transition for the TOF to begin counting time intervals.

Determining the Resolution of the Timer

Timers count time intervals. The resolution (or time base) of the timer determines the amount of time in each interval. For example, a TON with a resolution of 10 ms counts the number of 10-ms intervals that elapse after the TON is enabled: a count of 50 on a 10-ms timer represents 500 ms. The SIMATIC timers are available in three resolutions: 1 ms, 10 ms, and 100 ms, the timer number determines the resolution of the timer.

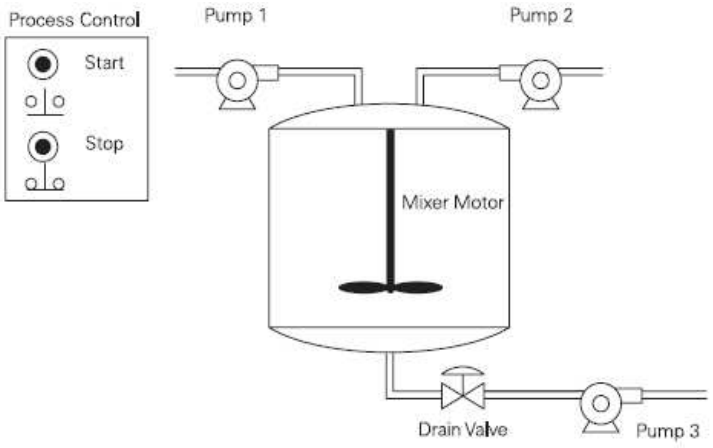
Timer Type	Resolution	Maximum Value	Timer Number
TON TOFF	1ms	32.767 seconds	T32,T96
	10ms	327.67 seconds	T33-T36 , T97 – T100
	100ms	3276.7 seconds	T37 – T63 , T101 – T255
TONR	1ms	32.767 seconds	T0 , T64
	10ms	327.67 seconds	T1-T4 , T65 – T68
	100ms	3276.7 seconds	T5-T31 , T69-T95

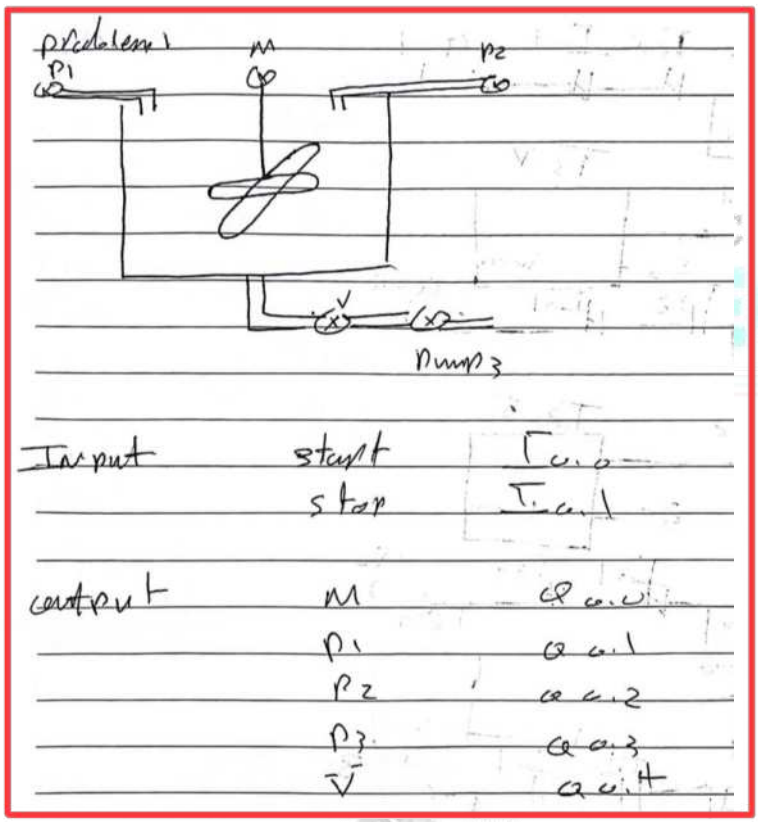
Problem 1

A tank will be filled with two chemicals, mixed, and then drained. When the Start Button is pressed, the program starts pump 1 runs for 5 seconds, filling the tank with the first chemical, then shuts off. The program then starts pump 2, runs for 3 seconds filling the tank with the second chemical. After 3 seconds pump 2 shuts off. The program starts the mixer motor, and mixes the two chemicals for 60 seconds. The program then opens the drain valve, and starts pump 3.

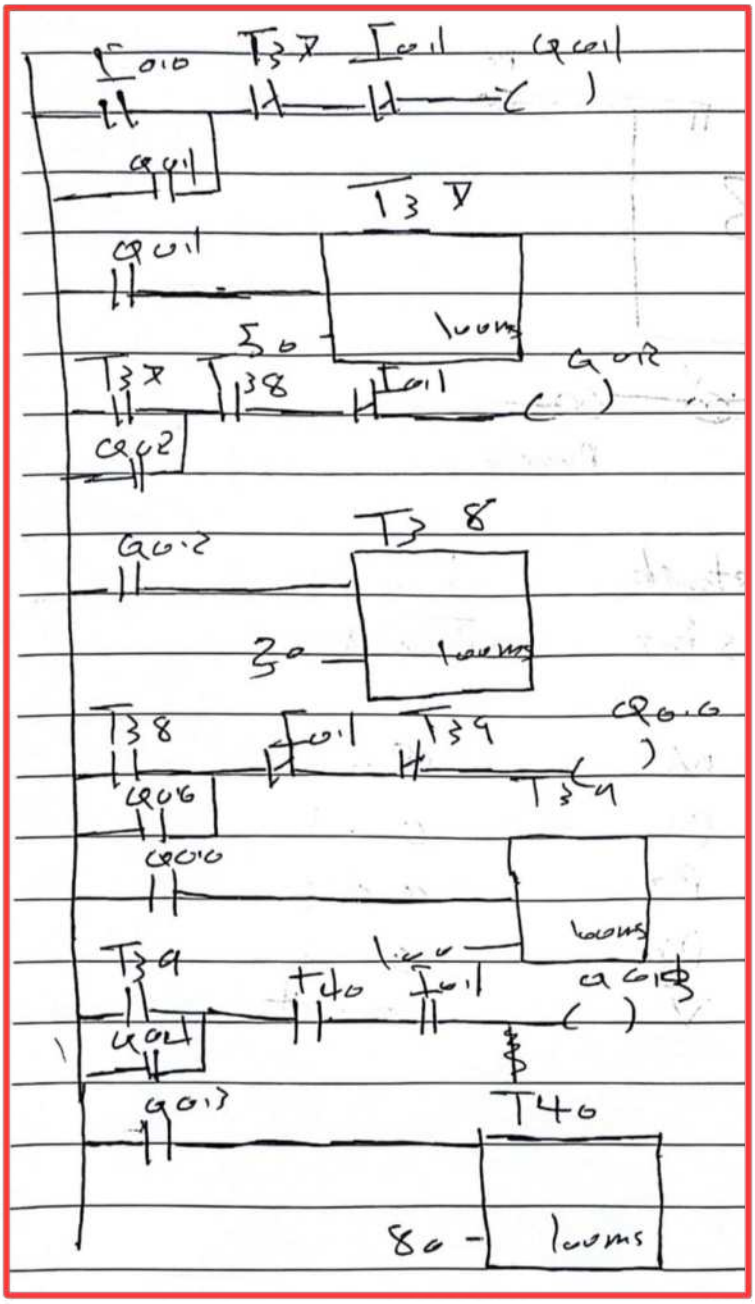
Pump 3 shuts off after 8 seconds and the process stops. A manual Stop switch is also provided to stop process at any time .

- Write a ladder diagram for S7-200 PLC to control the system
- Input / output table
- Write program in instruction list





I/O list



Ladder digramm

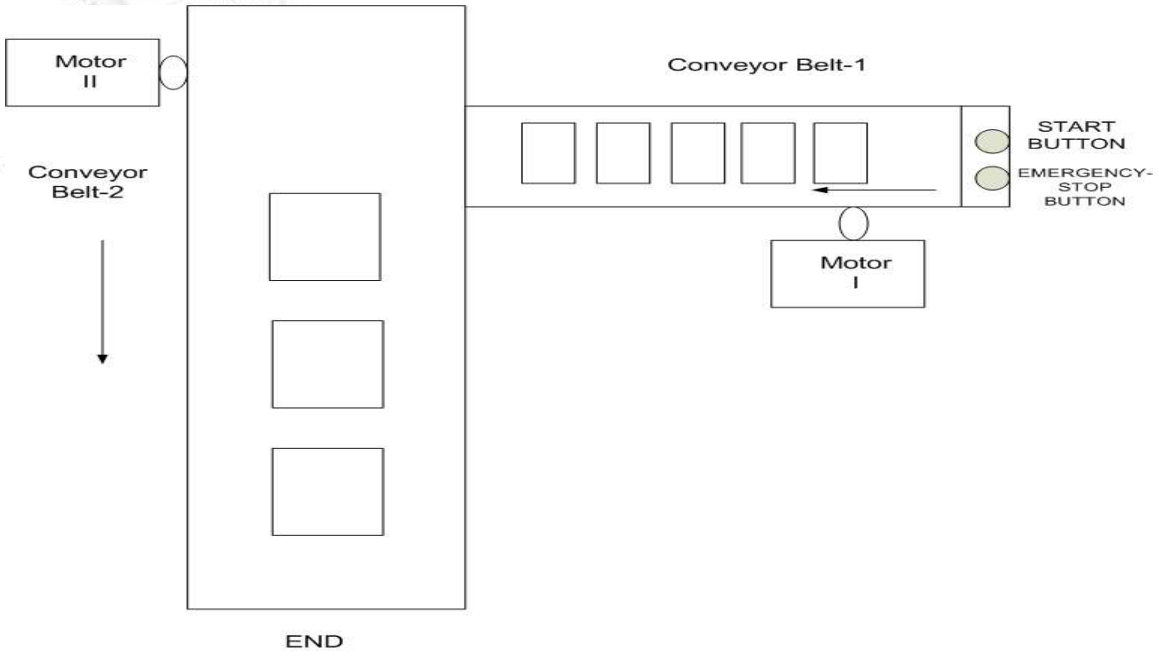
التجربة الرابعة

Timers Problems

Problem 1

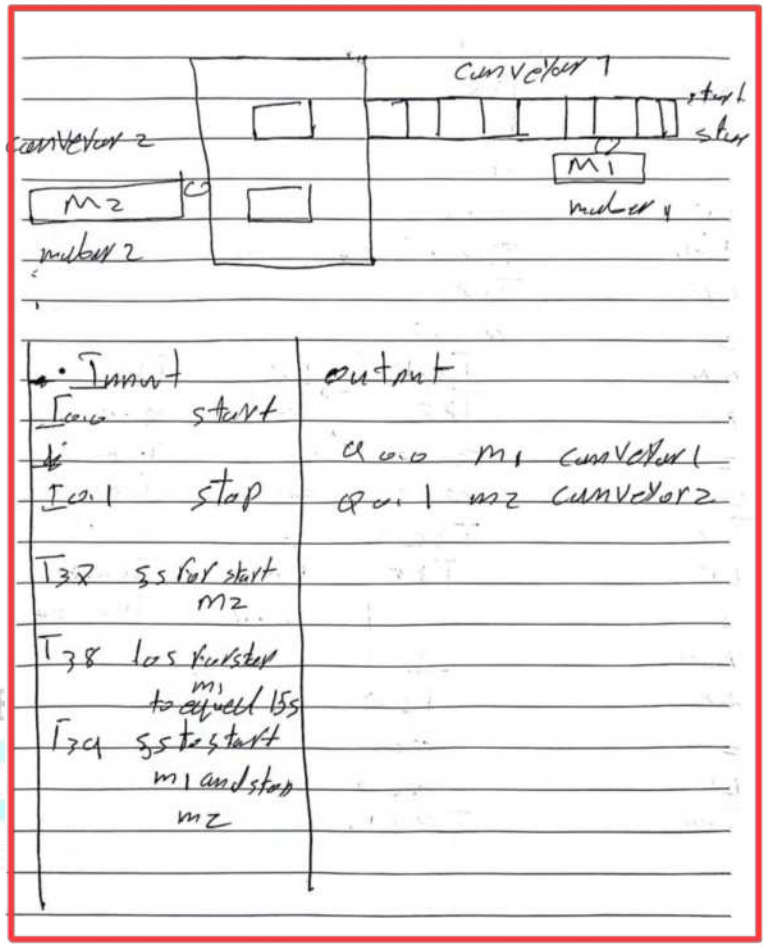
The system to be controlled by PLC consists of two belts. If the Start button is pressed, Conveyor Belt-1 will begin to run. After 5 seconds Conveyor Belt-2 will be active. After the whole system runs for 15 seconds, Conveyor Belt-1 will stop. Then Conveyor Belt-2 continues to move for 5 seconds and then it will stop, too. Also the system can be reset by the emergency-stop button at any time.

- Write a ladder diagram for S7-200 PLC to control the system
- Input / output table



```
LD    I0.6
O     Q0.1
AN    T37
AN    I0.1
=     Q0.1
LD    Q0.1
TON   T37, 50
LD    T37
O     Q0.2
AN    I0.1          =>  LD    T39
AN    T38               O     Q0.4
=     Q0.2              AN    T40
LD    Q0.2              AN    I0.1
TON   T38, 30           =     Q0.4
LD    T38               LD    Q0.3
O     Q0.1              TON   T40, 80
AN    I0.1
AN    T39
=     Q0.6
LD    Q0.0
TON   T39, 100
```

instruction list



input	output
I0.6 start	Q0.0 M1 conveyor 1
I0.1 stop	Q0.1 M2 conveyor 2
T37 5s for start M2	
T38 1os for conveyor M1 to adjust 15s	
T39 5s to start M1 and stop M2	

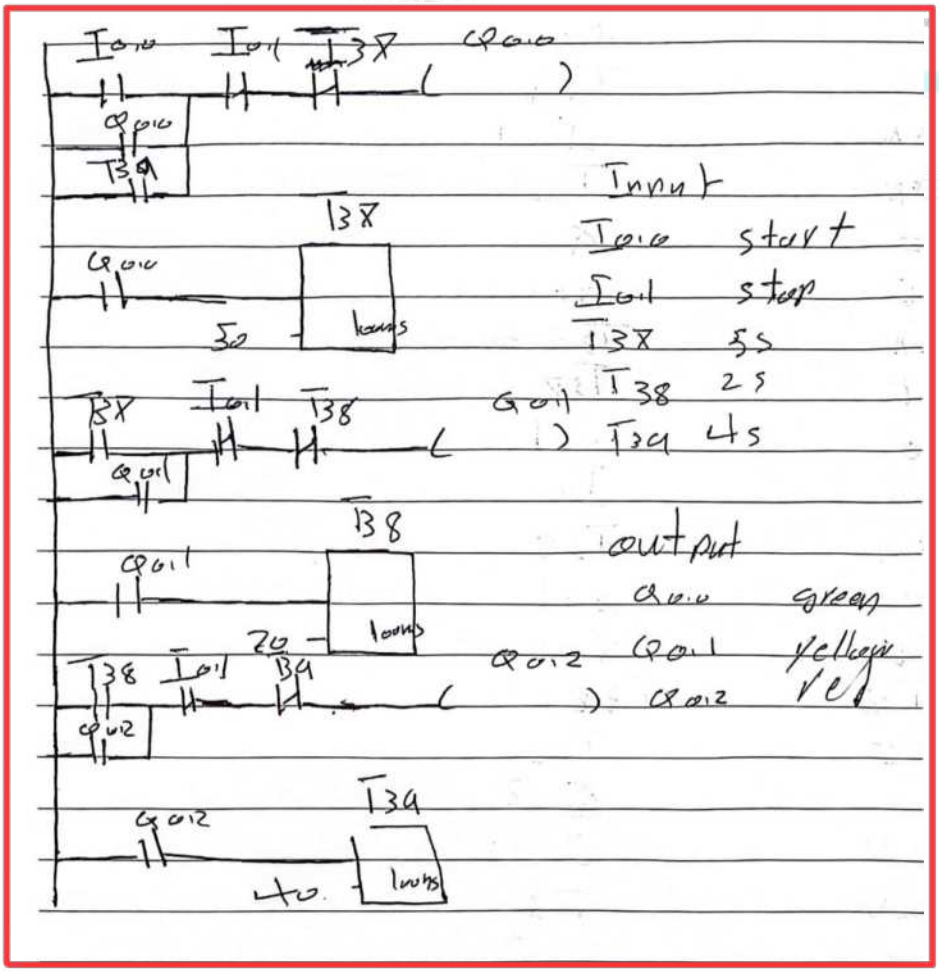
I/O list

بسواعدنا نبنيها
MECHATRONICS SPECTRUM TEAM

Problem 2

Control a traffic light system using plc with the following sequence:

1. The red light is on for 5 seconds
2. The red light turns off and the yellow light is turned on for 2 seconds
3. The yellow light turns off and the green light is turned on for 4 seconds
4. The red light turns on again



ladder diggram

```
LD      T0.01
O       Q0.0
O       T34
AN      I0.1
AN      T37
=       Q0.0
LD      Q0.0
TON     T37,30
LD      T37
O       Q0.1
AN      T0.1
AN      T38
=       Q0.1
LD      Q0.1
TON     T38,20
LD      T38
O       Q0.2
AN      I0.1
AN      T34
=       Q0.2
LD      Q0.2
TON     T34,40
```

instruction list

MECHATRONICS SPECTRUM TEAM

بسواعدنا نبنيها

التجربة الخامسة

Communication PLC Siemens S7-200 to PC and Micro-Win

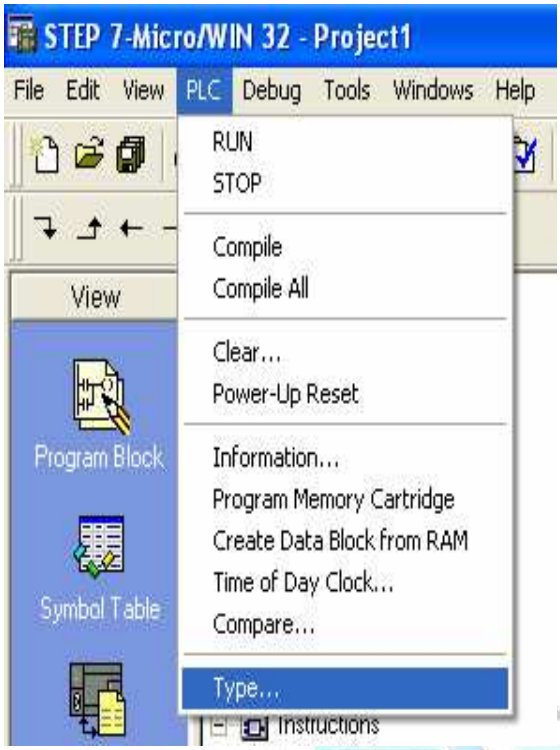
Communication PLC to PC

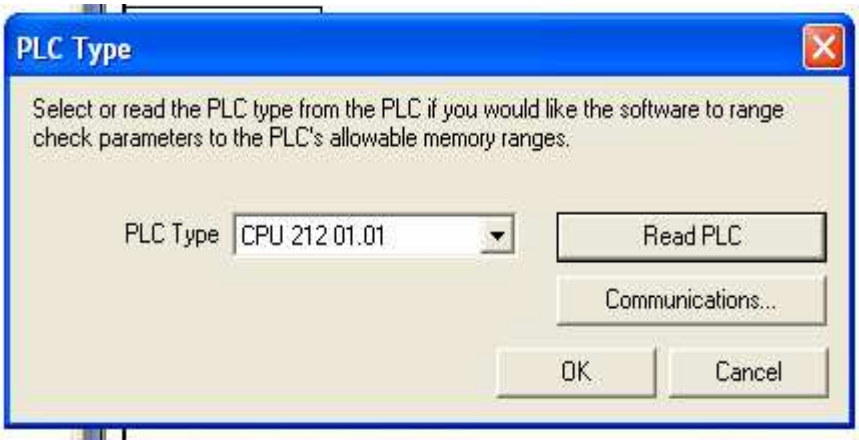
1. Siemens PLC used in this program: S7-200 CPU212

a) Connect part number PC/PPI cable from PC to Siemens S7-200.

2. Launch Siemens programming software.
3. Select PLC – Type.
4. Next, click on the Read PLC button to read your PLC type.
5. Click on the Communications button and the settings should appear

MECHATRONICS SPECTRUM TEAM

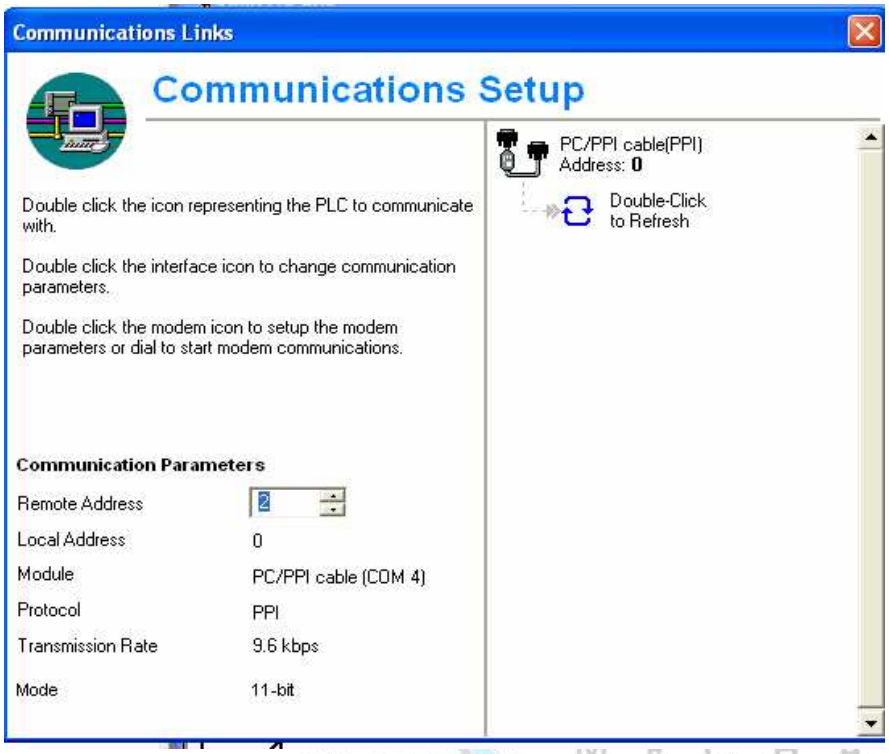




6. Assign the Remote Address (in this testing, #2 is assigned).

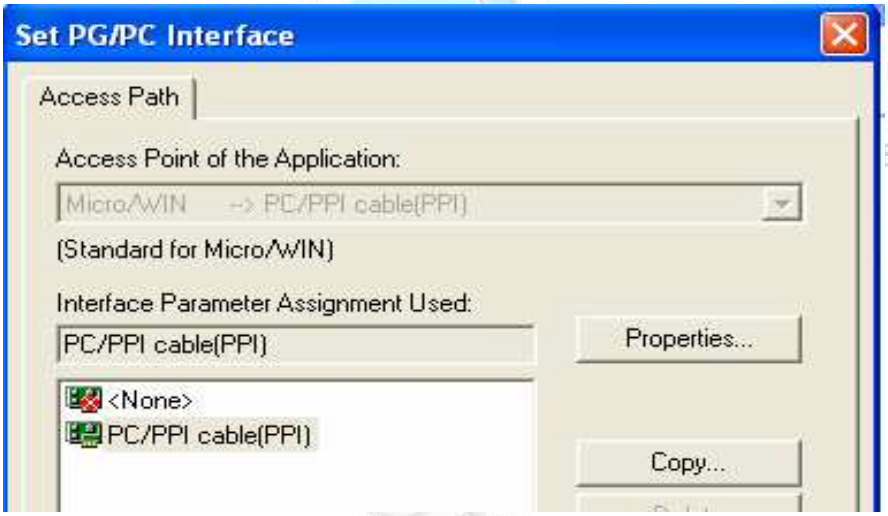


7. Double click



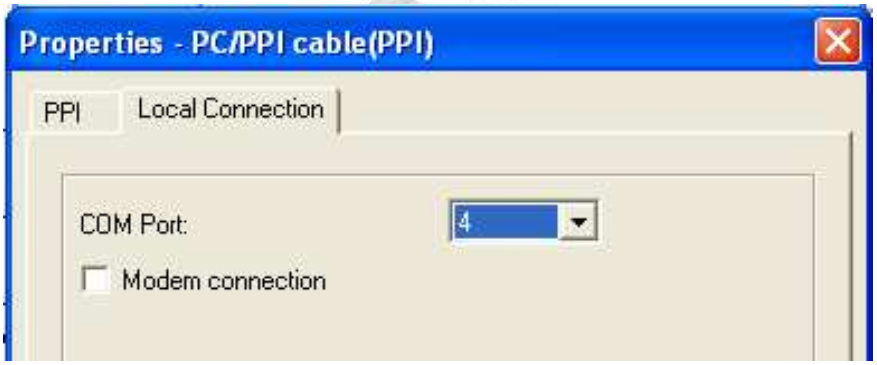
8. In the Access Path, select PC/PPI cable (PPI).

a) Click the Properties button.



9. Local Connection:

a) Assign the COM Port. Check your COM port before assigning the number.



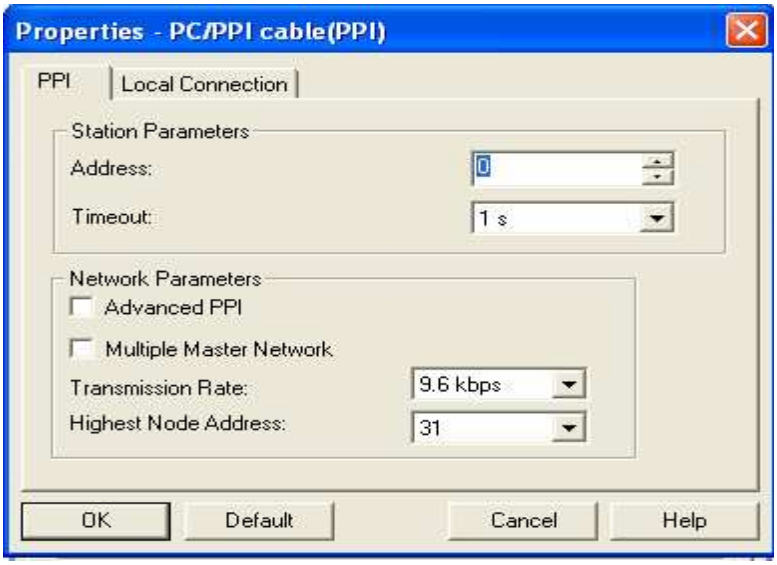
10. PPI :

a) Assign the station address (#0 is assigned for this test)

b) Set the transmission rate at 9.6kbps

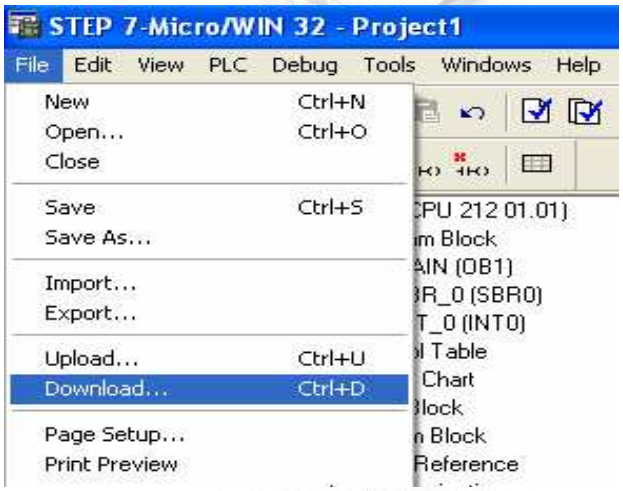
c) Highest node address is 31

d) Click the OK button to exit from the Properties dialogue box

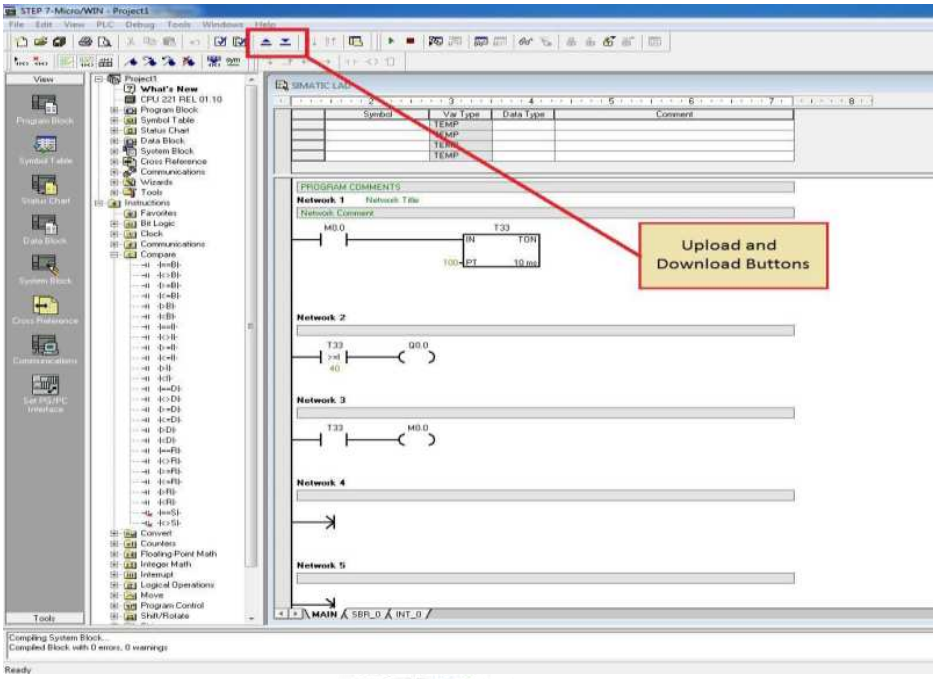


Downloading the Program to PLC

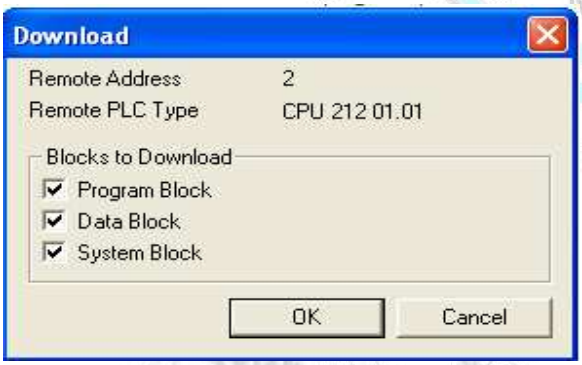
- Click the Download icon on the toolbar or select the File > Download menu command to download the program



- Click OK on opened window to download the elements of the program to the S7-200.



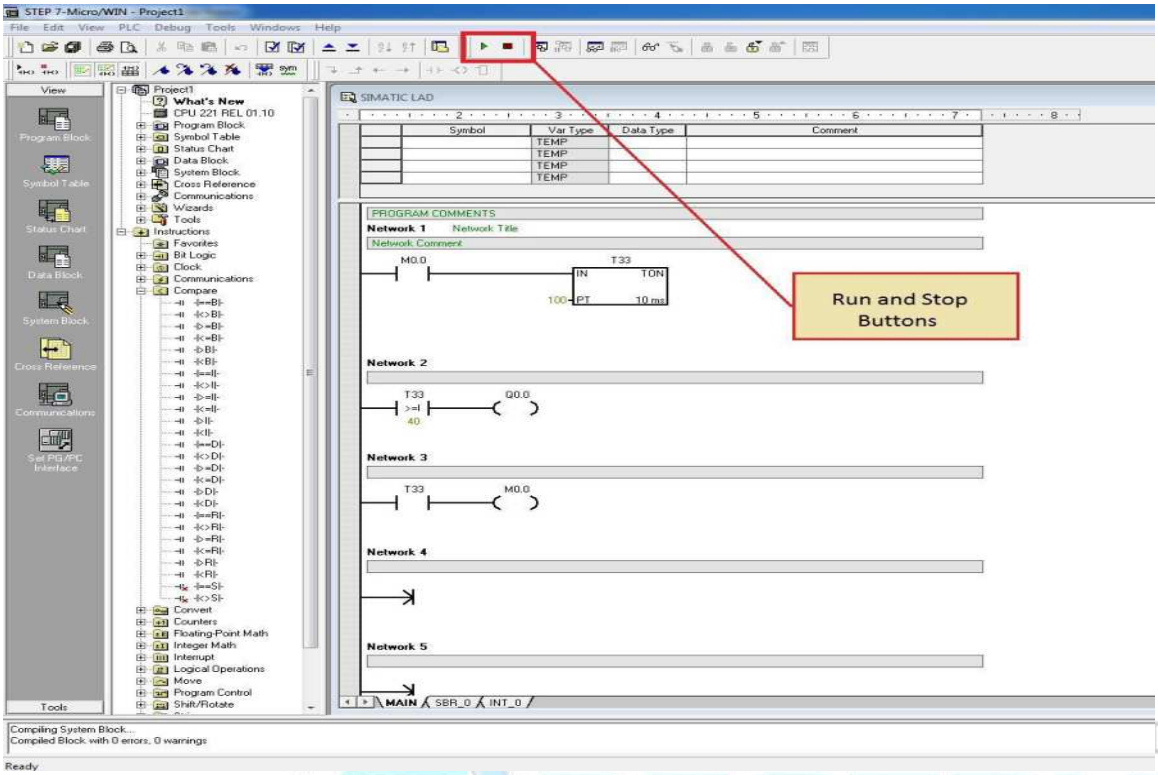
In the Download dialog box, follow all settings as shown in the image below and then click the OK button to download.



Running the PLC

For STEP 7--Micro/WIN to place the S7-200 CPU in RUN mode, the mode switch of the S7-200 must be set to TERM or RUN. When you place the S7-200 in RUN mode, the S7-200 executes the program:

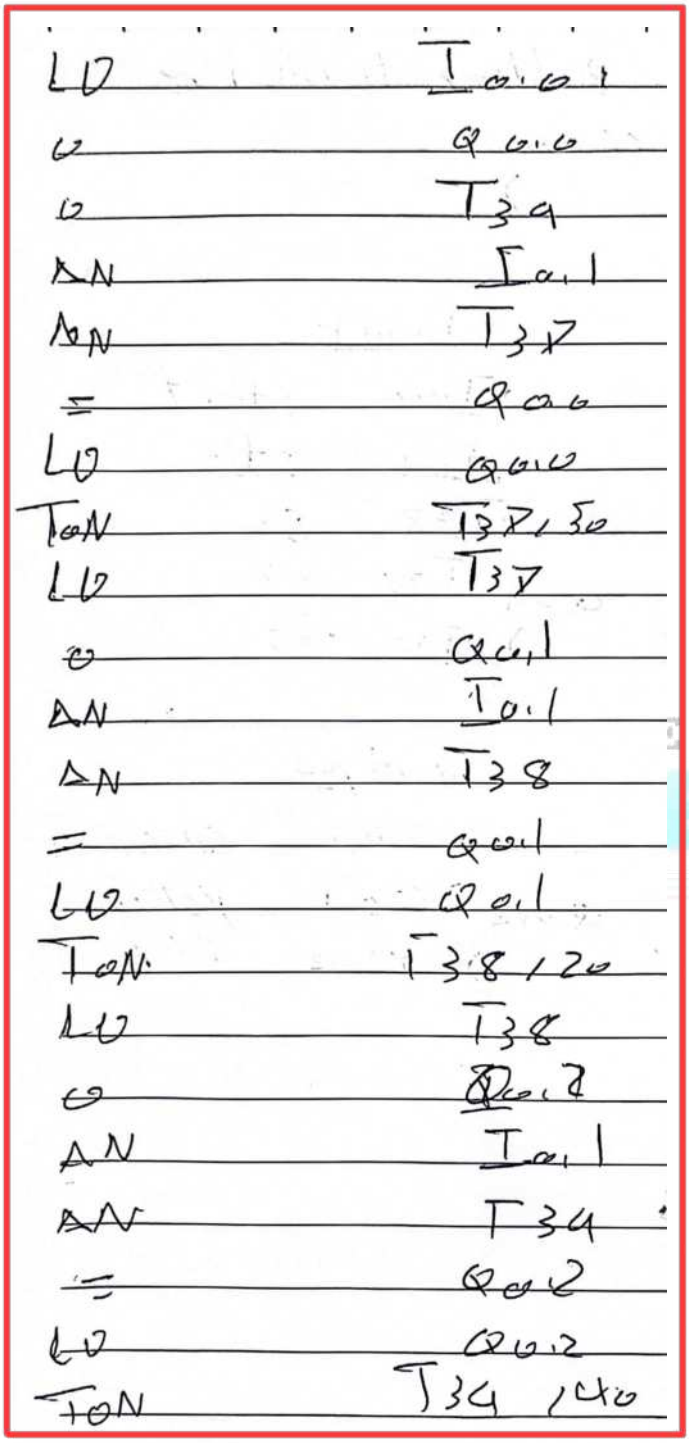
- Click the RUN icon on the toolbar or select the PLC > RUN menu command
- Click OK to change the operating mode of the S7-200



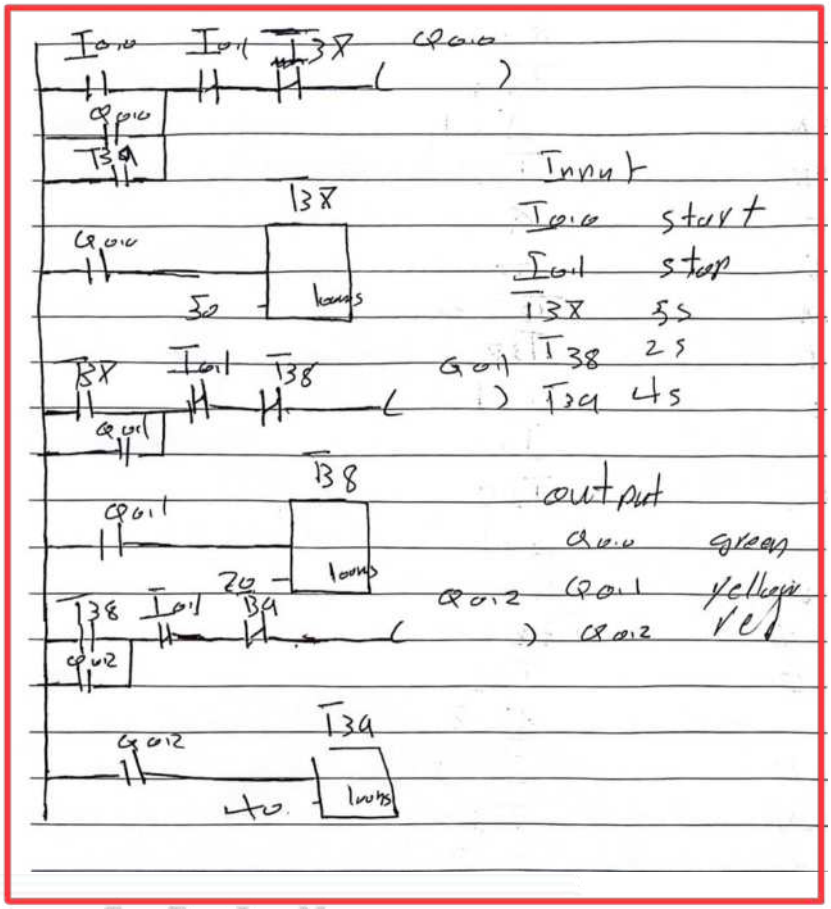
Problem 1

Control a traffic light system using plc with the following sequence:

5. The red light is on for a 5 seconds
6. The red light turns off and the yellow light is turned on for a 2 seconds
7. The yellow light turns off and the green light is turned on for a 4 seconds
8. The red light turns on again



instruction list



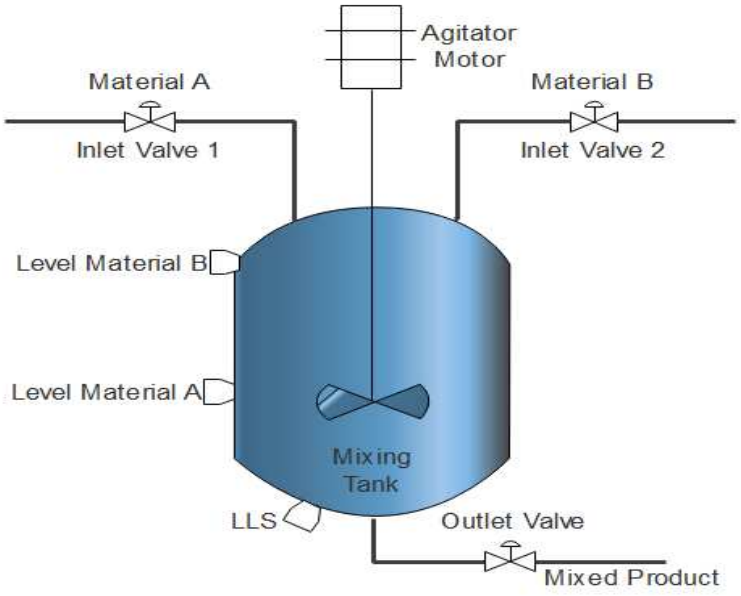
ladder diagram

Problem 2

Problem Description

Material A and Material B are collected in a tank. These materials are mixed for a while. Mixed product is then drained out through Outlet valve. Implement this in PLC using Ladder Logic programming language.

Problem Diagram

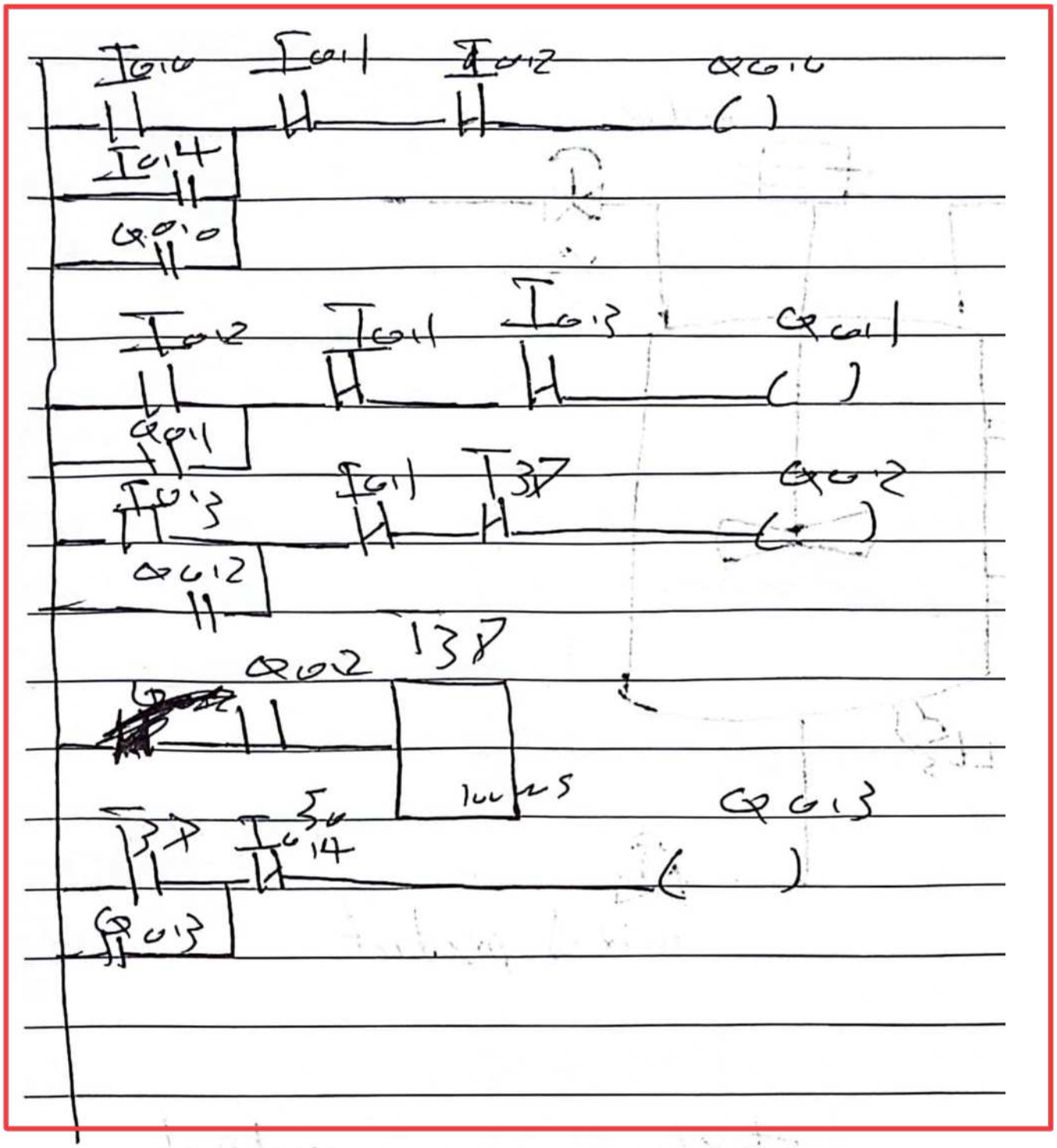


ATRONICS
CTRUM
TEAM

Input			
I0.0	start	output	
I0.1	stop	Q0.0	V1
~~I0.2~~		Q0.1	V2
I0.2	level A	Q0.3	Motor
I0.3	level B	Q0.4	V3
I0.4	LLS		mixed product

I/O list

بسواعدنا نبنيها



ladder diagram

بسواعدنا نبنيها
MECHATRONICS SPECTRUM TEAM

مادة الفاينل



MECHATRONICS
SPECTRUM
TEAM

بسواعدنا نبنيها

التجربة السادسة

COUNTERS

Counter UP (CTU)

The Count Up instruction (CTU) counts up from the current value each time the count up (CU) input makes the transition from off to on. When the current value Cxx is greater than or equal to the preset value PV, the counter bit Cxx turns on. The counter is reset when the Reset (R) input turns on, or when the Reset instruction is executed. The counter stops counting when it reaches the maximum value (32,767).

Counter Down (CTD)

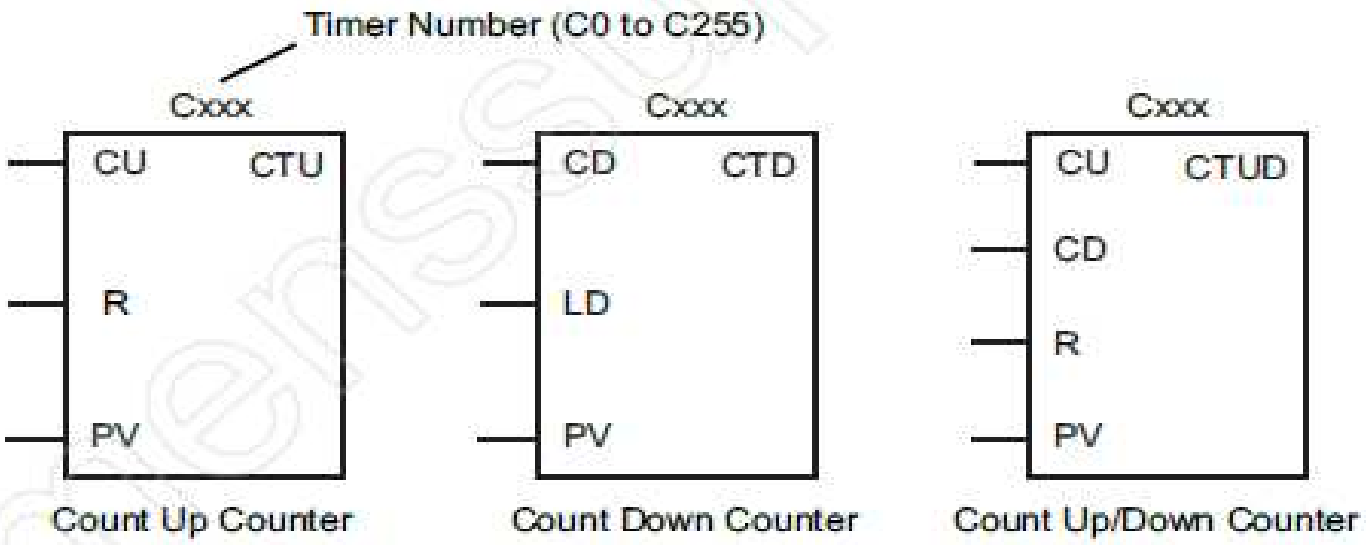
The Count Down instruction (CTD) counts down from the current value of that counter each time the count down (CD) input makes the transition from off to on. When the current value Cxx is equal to 0, the counter bit Cxx turns on

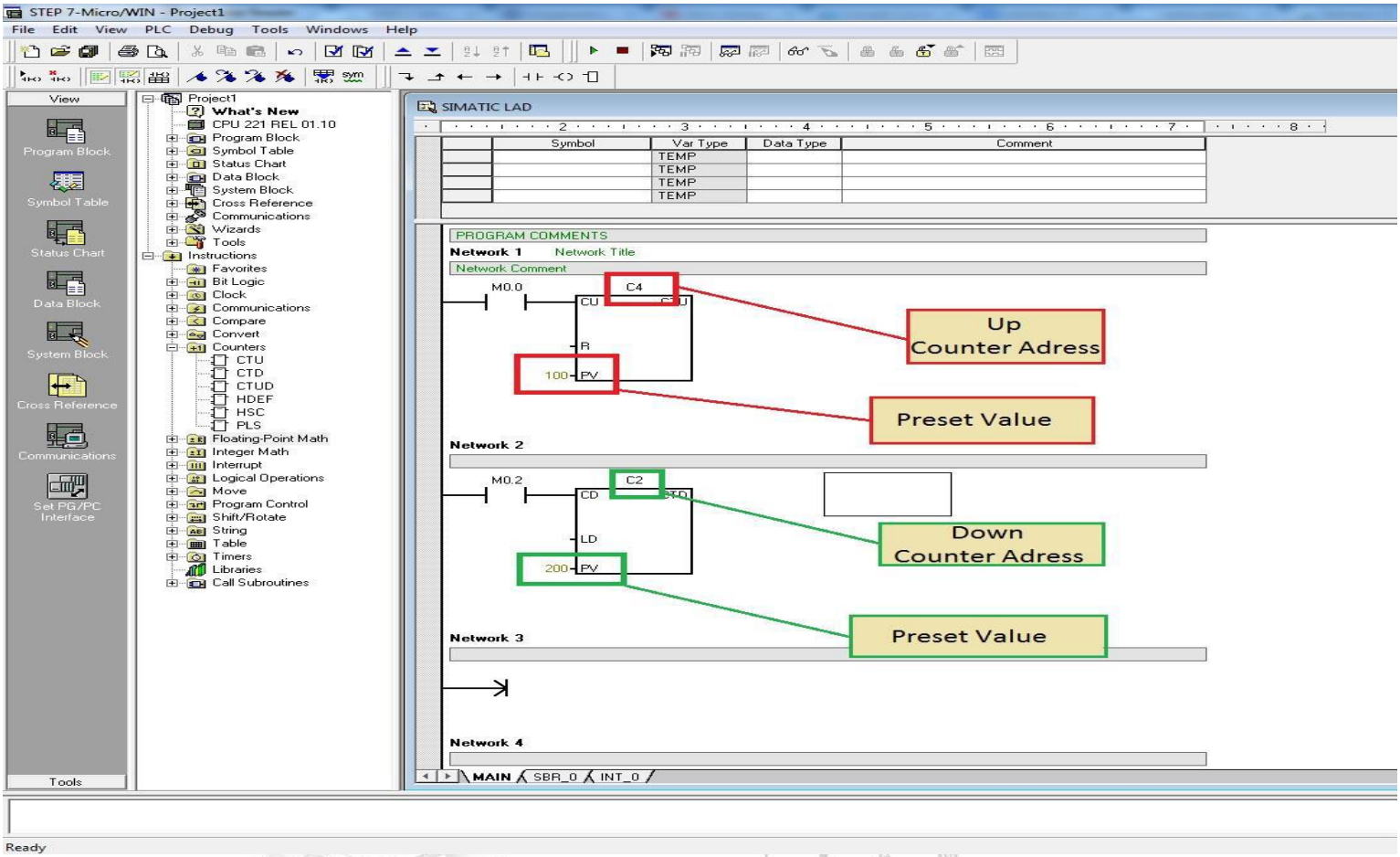
The counter resets the counter bit Cxx and loads the current value with the preset value PV when the load input LD turns on. The counter stops upon reaching zero, and the counter bit Cxx turns on.

R input for CTU and LD input for CTD are reset inputs of counters. You can reset counters by using this inputs.

Counter UO/Down (CTUD)

The counter UP/Down (CTUD) counts up by one each time the count up (CU) input transitions from off to on and count sown by one each time the count down (CD) input transitions from off to on. When the accumulated count equals the preset value (PV), the counter bit turns on. When the reset input (R) turns on or when a Reset instruction is executed, the accumulated count resets to zero and the counter bit turns off.



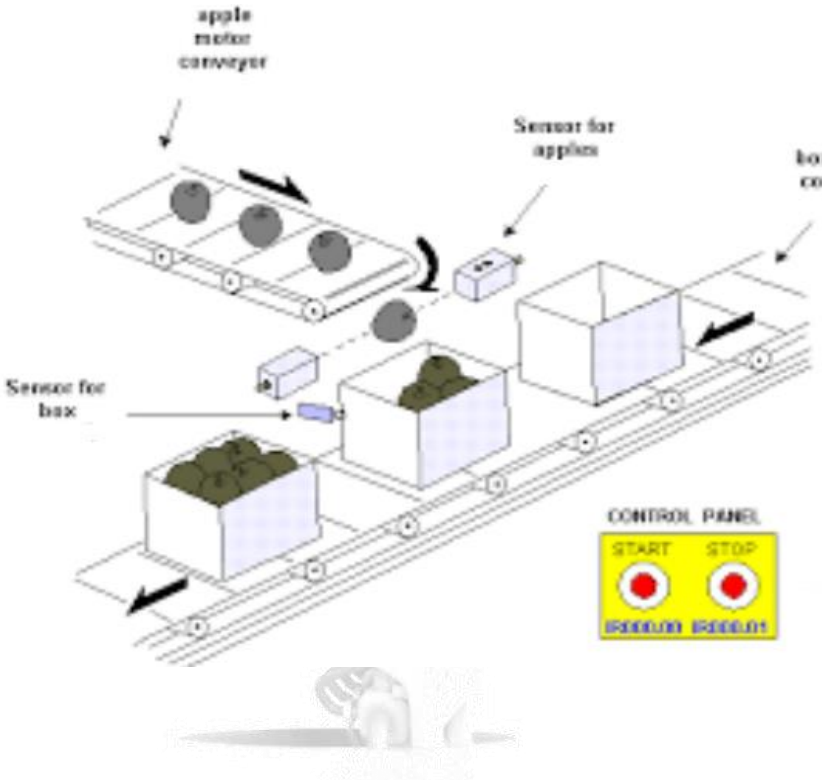


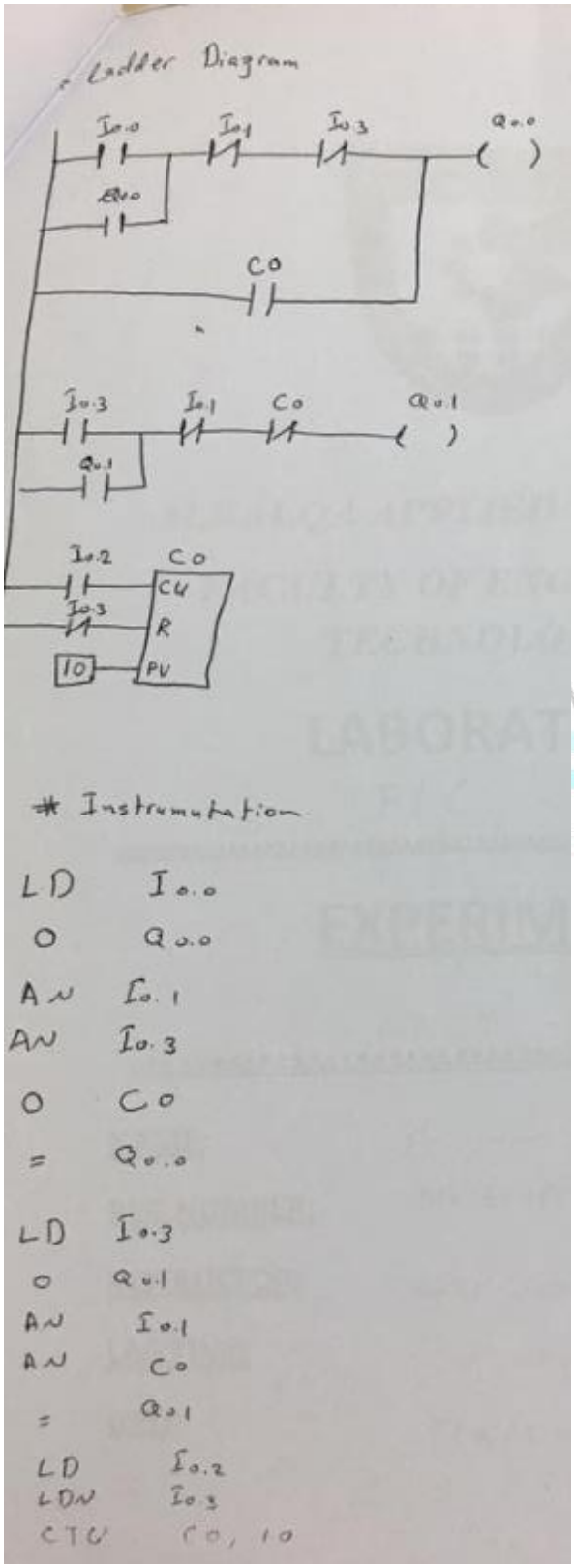
Type	Operation	Counter Bit	Power Cycle/First Scan
CTU	CU increments the current value. Current value continues to increment until it reaches 32,767.	The counter bit turns on when: Current value >= Preset	Counter bit is off. Current value can be retained.[1]
CTUD	CU increments the current value. CD decrements the current value. Current value continues to increment or decrement until the counter is reset.	The counter bit turns on when: Current value >= Preset	Counter bit is off. Current value can be retained.[1]
CTD	CD decrements the current value until the current value reaches 0.	The counter bit turns on when: Current value = 0	Counter bit is off. Current value can be retained.[1]

Operations of the Counter Instructions

Problem 1

In this system by pushing start key you activate the power for the system (resetting depends only on a stop key). When started, motor of conveyer for boxes is activated. The conveyer takes a box up to the limit switch, when a box is detected, the motor stops then a conveyer with apples starts moving. Presence of the box allows counter to count 10 apples through a sensor used for apples. Operations repeat until stop key is pressed.





instruction list

Inputs		Outputs	
start	I 0.0	Q 0.0	M1
off	I 0.1	Q 0.1	M2
S1	I 0.2		
LS	I 0.3		

I/O list

MECHATRONICS SPECTRUM TEAM

بسواعدنا نبنيها
MECHATRONICS SPECTRUM TEAM

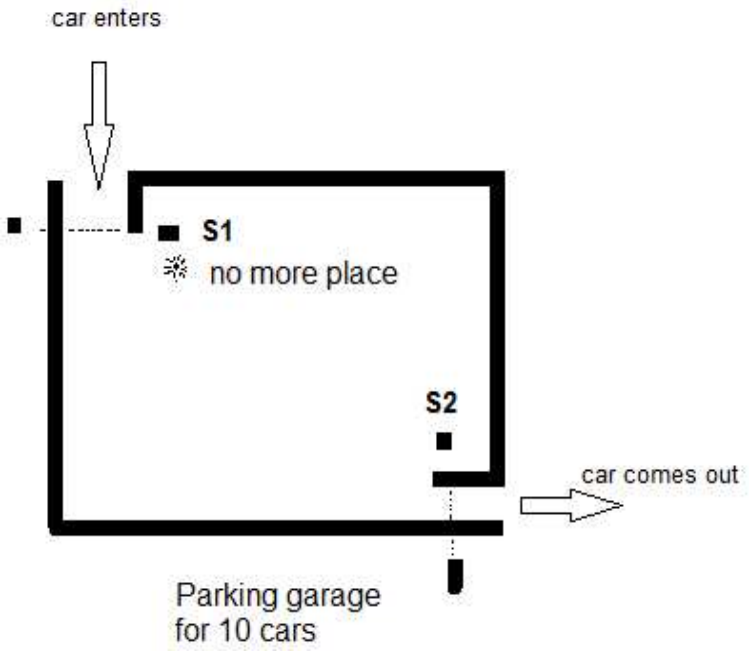
التجربة السابعة

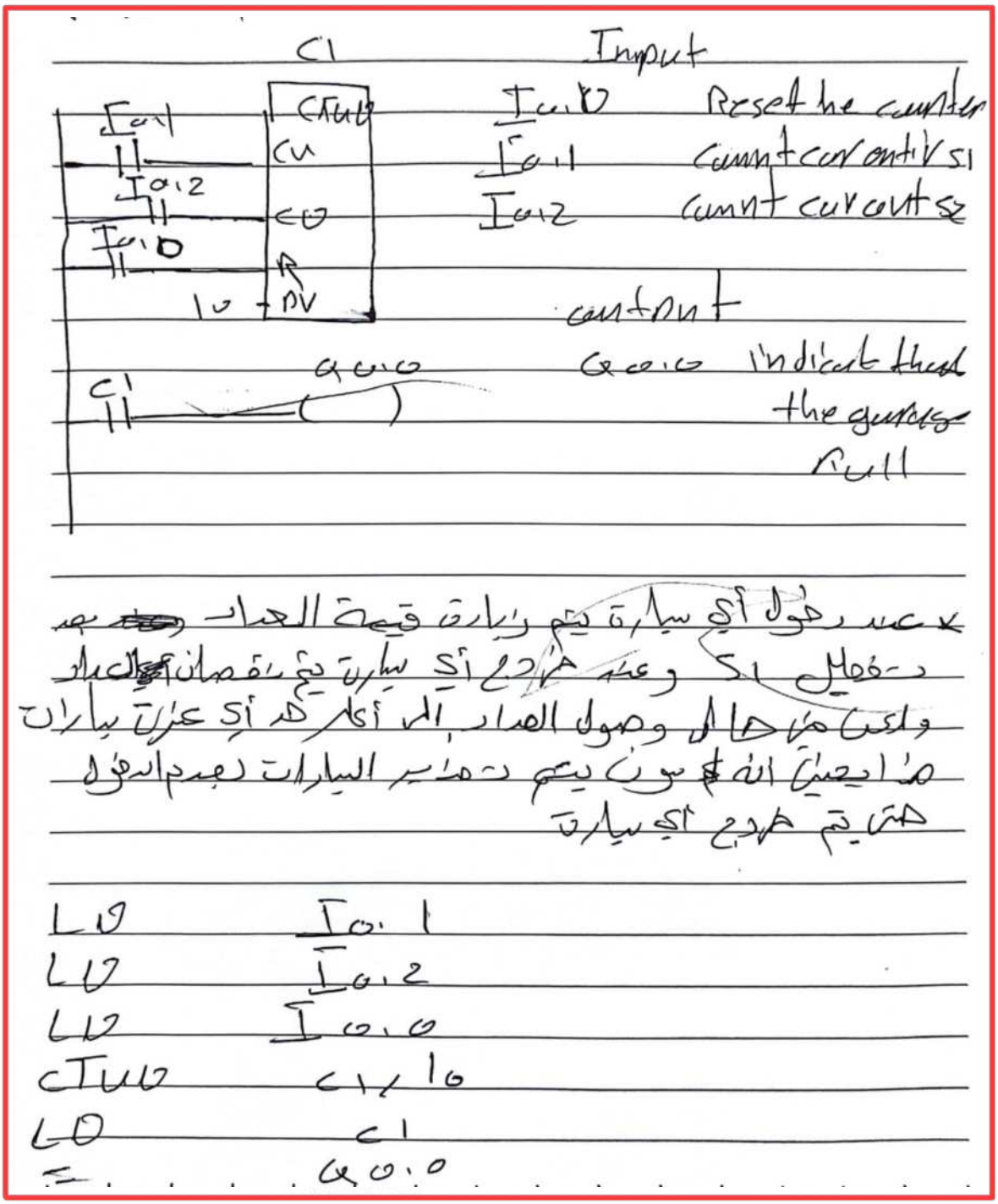
COUNTERS Problems

Problem 1

We are dealing with a simple system that can control 10 cars at the maximum. Each time a car enters, PLC automatically adds it to a total sum of other cars found in the garage. Each car that comes out will automatically be taken off. When 10 cars park, a signal will turn on signalizing that a garage is full and notifying other drivers not to enter because there is no space available.

- Construct a LAD for this system.
- Construct input/output table.
- Construct a instruction list for this system





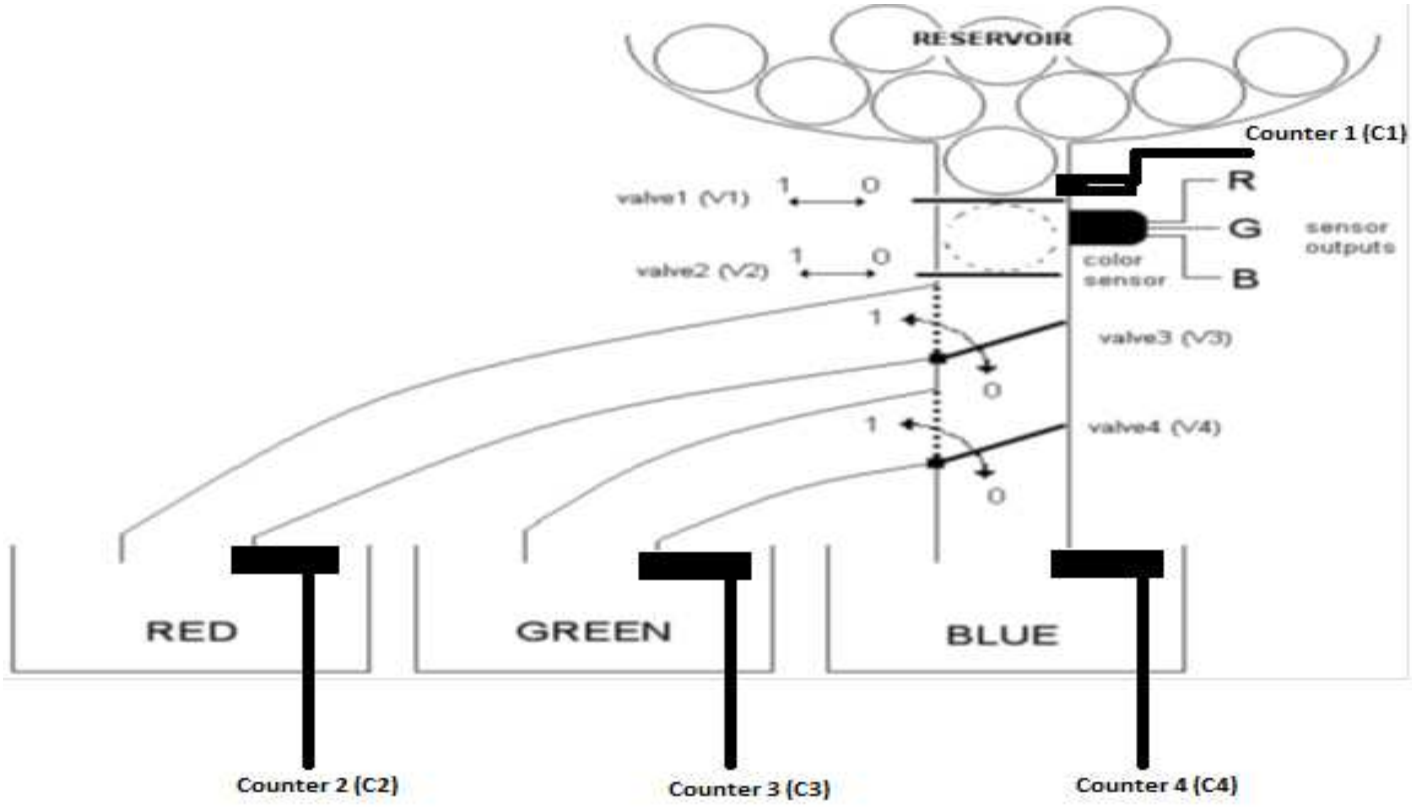
عند دخول أي سيارة يتم زيادة قيمة العداد بمقدار 1 وعند خروج أي سيارة يتم إنقاص العداد ولكن في حال وصول العداد إلى الرقم 10 أي عدد السيارات المسموح بها في الجراج لن يتم تسجيل السيارات بعد ذلك حتى يتم خروج أي سيارة

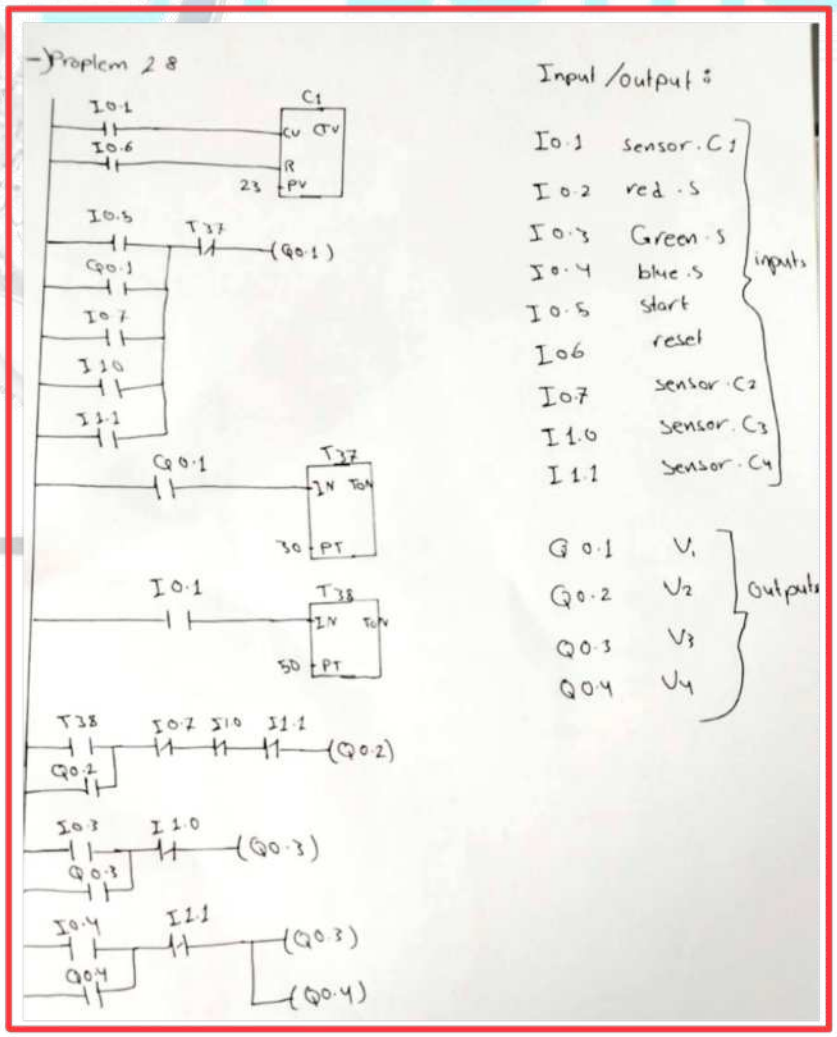
instruction list + ladder

Problem 2

The Figure below shows a system which separate three different color balls which are mixed in a reservoir and then fill three different labeled with color names bins. The total number of balls is 23 so that you have to count number of felt balls from reservoir by using counter 1 sensor to stop the system when reservoir becomes empty. A color sensor is used to recognize the color of the ball. If there is a red ball in front of the sensor, then " R " output of the sensor becomes 1. "G" output becomes 1, if the ball is green and "B" becomes 1, when the ball is blue. Four valves are used to run the process properly. Directions and input signals of the valves are shown in The Figure. Valve1 (V1) will be opened for 3 seconds then valve 2 (V2) will wait for 5 seconds to sensor measurement can be achieved. After measurement if the ball is red valve 3 (V3) and valve 4 (V4) will be remained on close position. If the ball is green V3 will opened but V4 will remained as closed. Finally if the ball is blue V3 and V4 will be opened. At the end of process we should know the number of balls in each bin (Number of Green Balls, Number of Red Balls and Number of Blue Balls).

- Construct a LAD for this system.
- Construct input/output table.
- Construct a instruction list for this system





ladder + instruction list

التجربة الثامنة

PLC Wiring Input and Output

INPUTS AND OUTPUTS

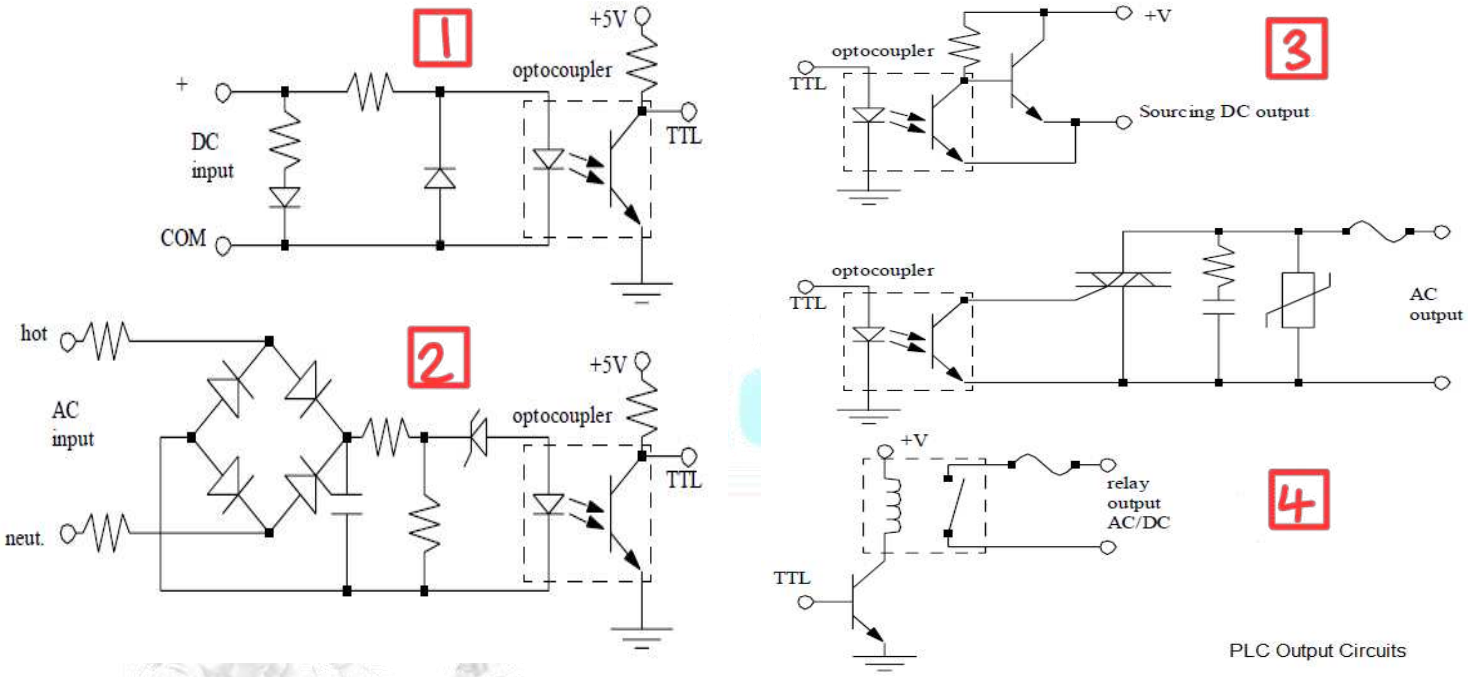
Inputs and outputs from a PLC are necessary to monitor and control a process. Both inputs and outputs can be categorized into two basic types: logical or continuous. Consider the example of a light bulb. If it can only be turned on or off, it is logical control.

If the light can be dimmed to different levels, it is continuous.

Outputs from PLCs are often relays, but they can also be solid state electronics such as transistors for DC outputs or Triacs for AC outputs. Continuous outputs require special output cards with digital to analog converters. Inputs come from sensors that translate physical phenomena into electrical signals.

PLC inputs must convert a variety of logic levels to the 5Vdc logic levels used on the data bus. This can be done with circuits similar to those shown below. Basically the circuits condition the input to drive an optocoupler. This electrically isolates the external electrical circuitry from the internal circuitry.

PLC outputs must convert the 5Vdc logic levels on the PLC data bus to external voltage levels. This can be done with circuits similar to those shown below. Basically the circuits use an optocoupler to switch external circuitry. This electrically isolates the external electrical circuitry from the internal circuitry.



PLC Output Circuits

INPUTS

In smaller PLCs the inputs are normally built in and are specified when purchasing the PLC. For larger PLCs the inputs are purchased as modules, or cards, with 8 or 16 inputs of the same type on each card.

Input devices such as (Manual Switches, Pushbuttons, Selector Switches, Thermostats, Thermocouple, Pressure Sensors, Liquid level detection, Flow meter, LVDT, Strain gauge, …etc)

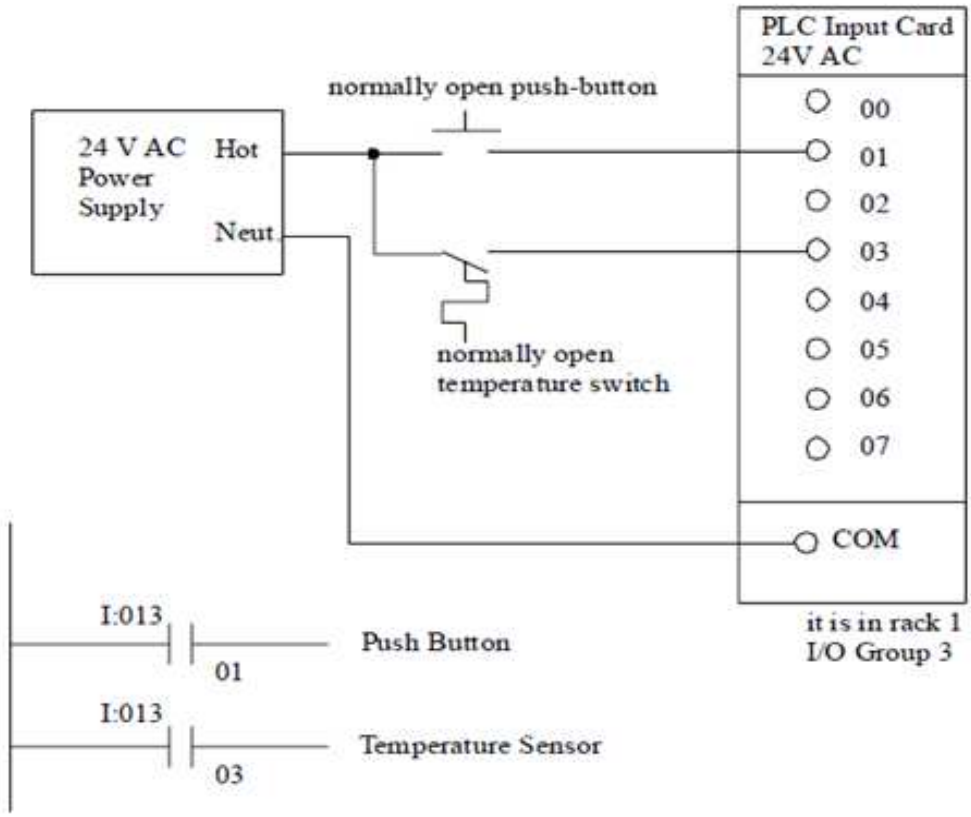
-Typical ranges for input voltages

12-24 Vdc / 100-120 Vac / 10-60 Vdc / 12-24 Vac/dc

5 Vdc (TTL) / 200-240 Vac / 48 Vdc / 24 Vac

PLC input cards rarely supply power, this means that an external power supply is needed to supply power for the inputs and sensors.

The example in Figure below shows how to connect an AC input card



An AC Input and Ladder Logic

In the example there are two inputs, one is a normally open push button, and the second is a temperature switch, or thermal relay. Both of the switches are powered by the hot output of the 24Vac power supply this is like the positive terminal on a DC supply. Power is supplied to the left side of both of the switches. When the switches are open there is no voltage passed to the input card. If either of the switches are closed power will be supplied to the input card.

Common (reference voltage) is the neutral on the power supply. In effect we have chosen this to be our 0V reference, and all other voltages are measured relative to it. If we had a second power supply, we would also need to connect the neutral so that both neutrals would be connected to the same common.

OUTPUTS

External power supplies are connected to the output card and the card will switch the power on or off for each output.

Output devices such as (Contactors and Relay, Motors, Pumps, Indication Lamps, Solenoid Vaves, Directional Control Valves)

-Typical output voltages are

120 Vac / 24 Vdc / 12-48 Vac /

12-48 Vdc / 5Vdc (TTL) / 230 Vac

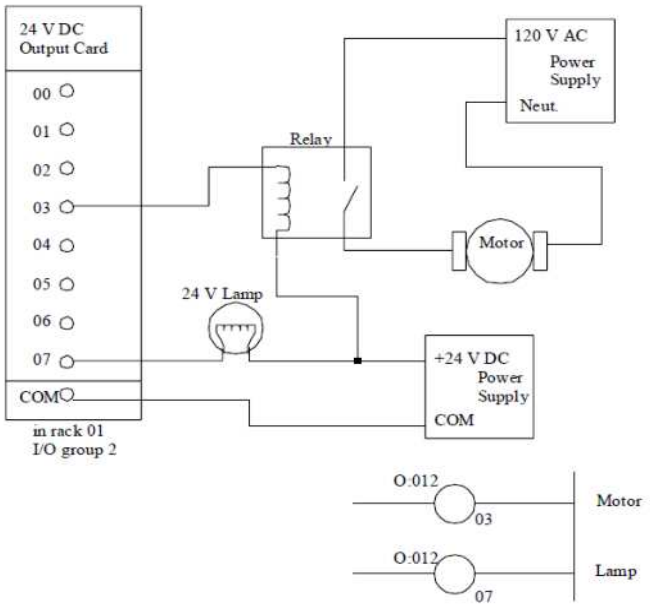
Relays are the most flexible output devices. They are capable of switching both AC and DC outputs. Transistors are limited to DC outputs, and Triacs are limited to AC outputs. Transistor and triac outputs are called switched outputs.

Sourcing and sinking inputs are also popular. This output method dictates that a device does not supply any power.

Sinking : When active the output allows current to flow to a common ground. This is best selected when different voltages are supplied.

Sourcing : When active, current flows from a supply, through the output device and to ground. This method is best used when all devices use a single supply voltage. This is also referred to as NPN (sinking) and PNP (sourcing). PNP is more popular.

The output card shown in figure below is an example of a 24Vdc output card that has a shared common. This type of output card would typically use transistors for the outputs.

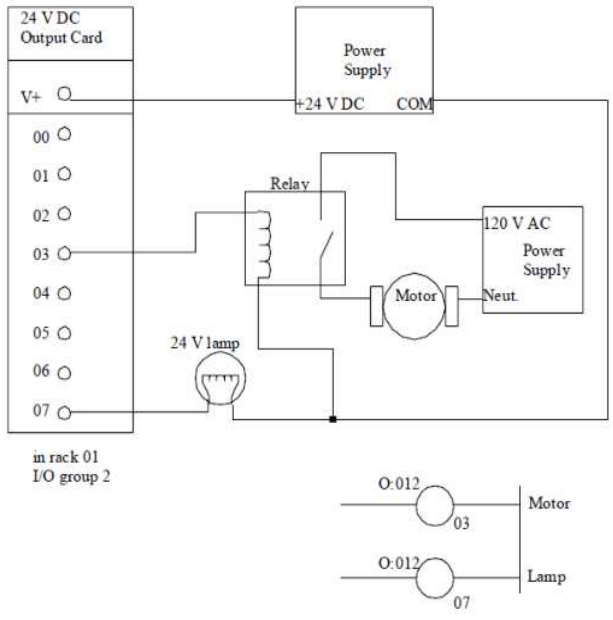


An Example of a 24Vdc Output Card (Sinking)

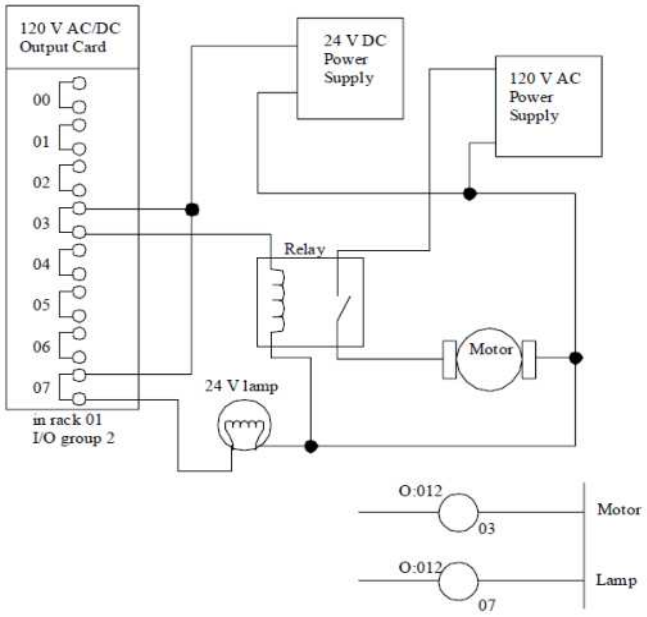
In this example the outputs are connected to a low current light bulb (lamp) and a relay coil. Consider the circuit through the lamp, starting at the 24Vdc supply. When the output 07 is on, current can flow in 07 to the COM, thus completing the circuit, and allowing the light to turn on. If the output is off the current cannot flow, and the light will not turn on

MECHATRONICS SPECTRUM

The output 03 for the relay is connected in a similar way. When the output 03 is on, current will flow through the relay coil to close the contacts and supply 120Vac to the motor. This card could have many different voltages applied from different sources, but all the power supplies would need a single shared common.



An Example of a 24Vdc Output Card With a Voltage Input (Sourcing)

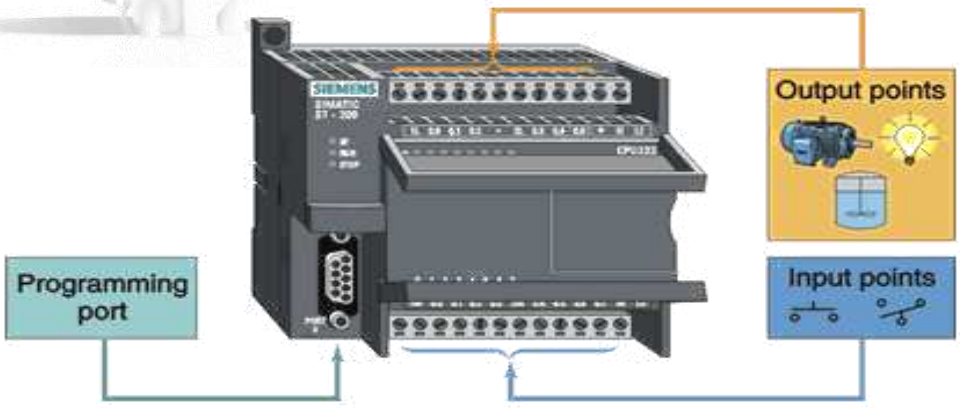


An Example of a Relay Output Card

Analog I/O

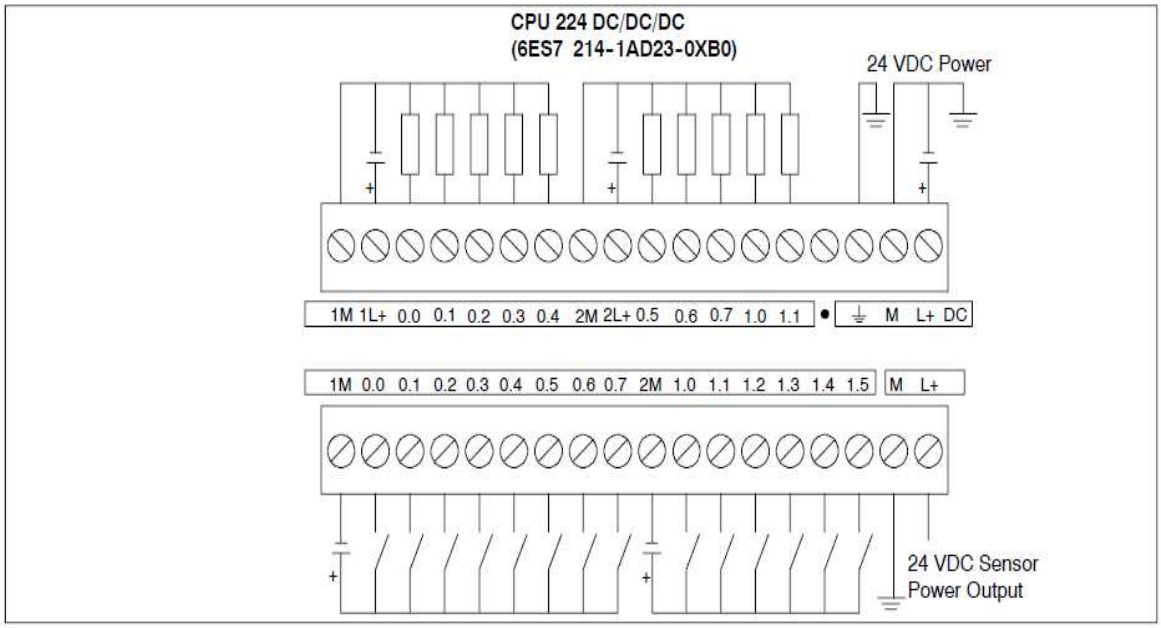
In the early days of programmable logic controllers, processor speed and memory were too limited to support anything but discrete (on/off) control functions. Modern PLC technology, though, is powerful enough to support the measurement, processing, and output of analog (continuously variable) signals. Thus, in order to interface with an analog sensor or control device, some “translation” is necessary between the analog and digital worlds. Inside every analog input module is an ADC, or Analog-to-Digital Converter, circuit designed to convert an analog electrical signal into a multi-bit binary word. Conversely, every analog output module contains a DAC, or Digital-to-Analog Converter, circuit to convert the PLC’s digital command words into analog electrical quantities.

Typical analogue signals from (0-10 VDC) or (4-20 mA). They are used to represent changing values such as Voltage, Current, speed, temperature, weight, level, Thermocouple, RTD and Strain gauge

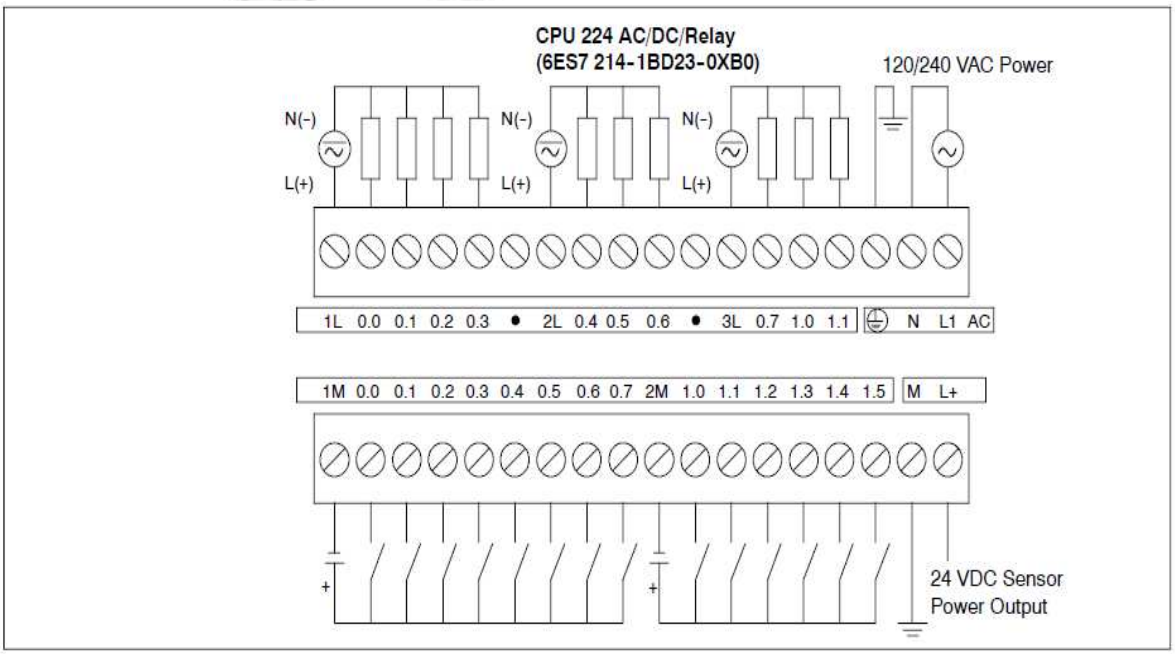


Wiring input/ output S7-200

CPU 224 DC/DC/DC



CPU 224 AC/DC/Relay



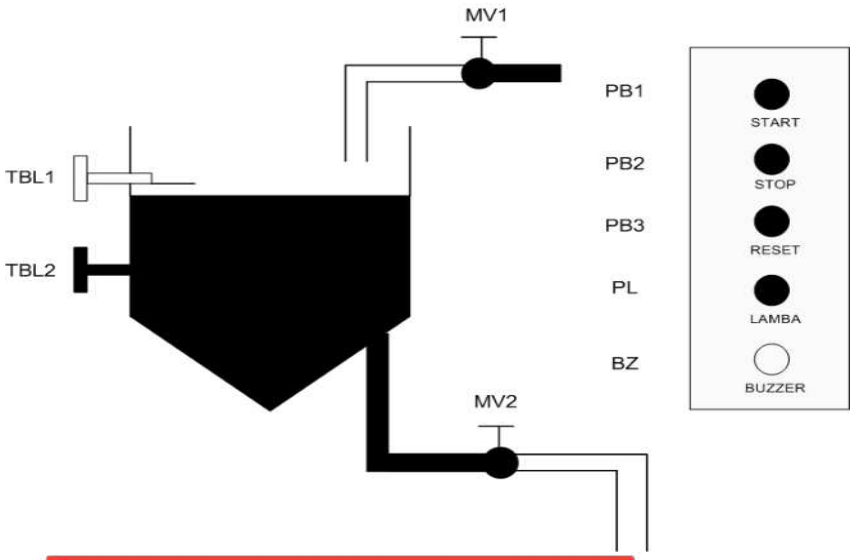
Problem

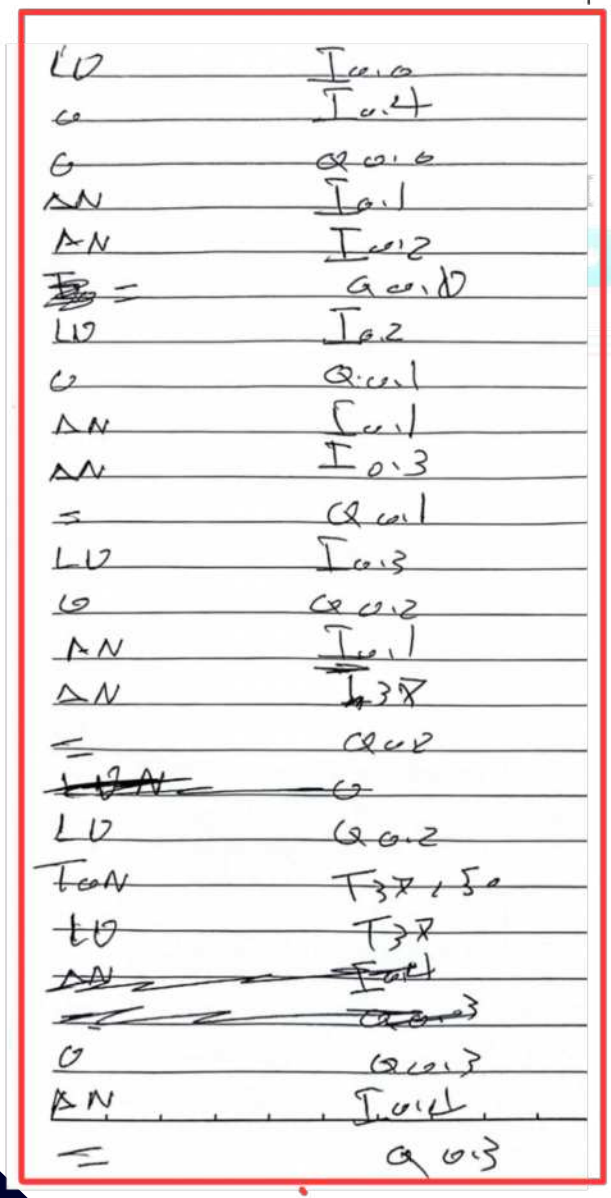
A PLC-controlled system operates as follows. If the Start button is pressed, MV1 will be opened and dye starts to fill the tank. At the same time mixing-motor begins to run. As the level of the dye passes TBL2 and reaches to TBL1, MV1 will be closed and mixing-motor will stop. Then MV2 will be opened and dye begins to run out of the container. After the level of the dye reaches to the below of TBL2, MV2 will be closed. This process is repeated for two (2) times, then the system will stop.

We can observe the stopping of the system with the lamp and the buzzer.

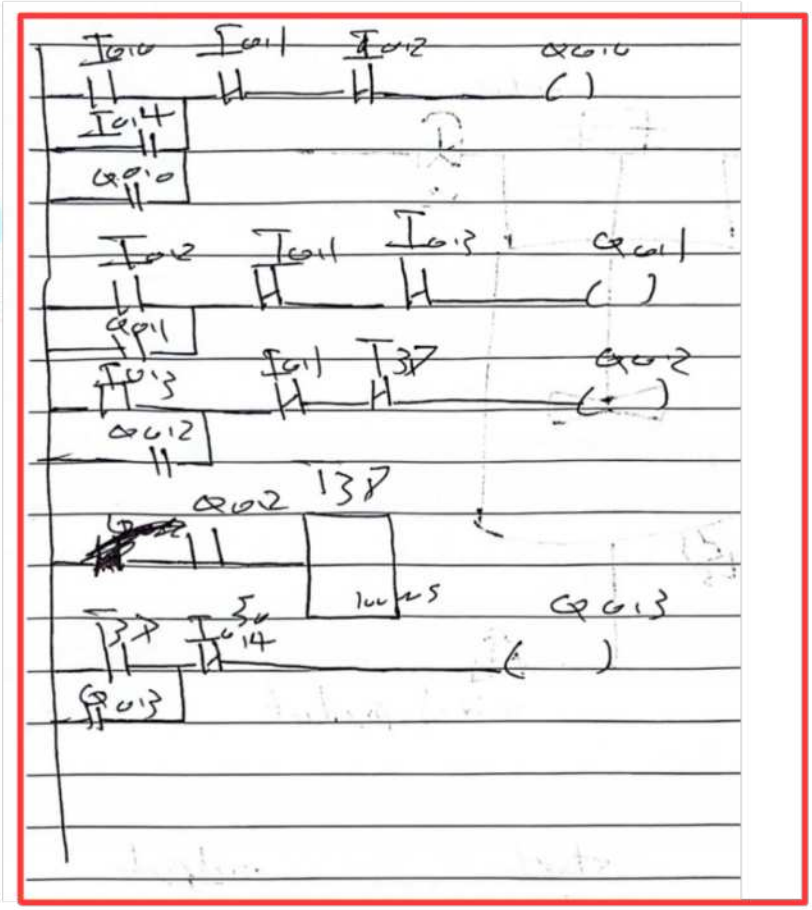
After the system stops, buzzer will continue to run for 2 sec and then it stops. But the lamp continues to light until the reset button is pressed. When the reset button is pressed, the system will be ready to begin the same process again.

- Construct a LAD for this system.
- Construct input/output table.
- Construct a instruction list for this system





instruction list



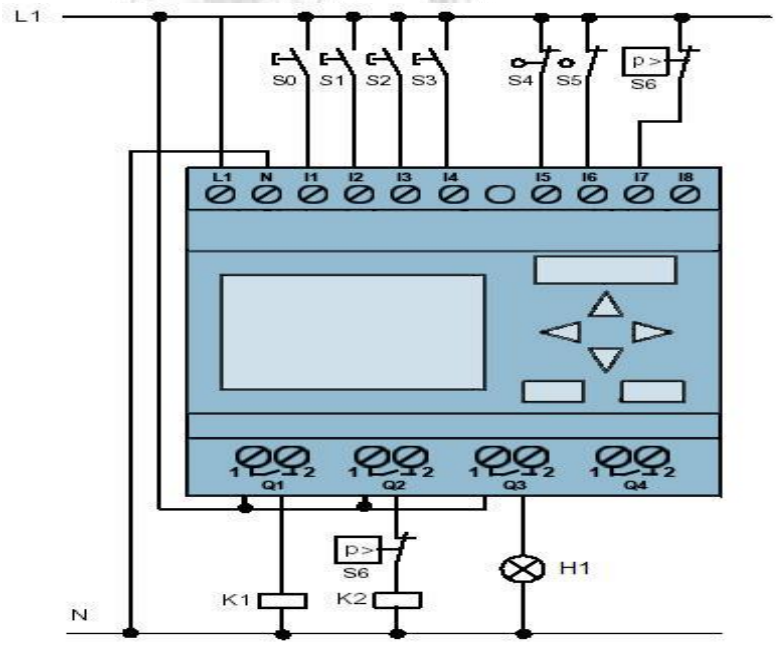
ladder diggram

التجربة التاسعة

LOGO LOGIC

The LOGO! logic module has an integrated controller with operator control and display unit. You can use the operator control and display unit of LOGO! to create and edit programs and execute system functions.

You can read in external programs from a program module via the Ethernet interface or a PC cable with the LOGO! Soft Comfort programming software. You can use LOGO! Soft Comfort not only to create a program but also to simulate your circuit on the computer and to output overview diagrams to a printer.



This PLC shows 8 inputs, 4 outputs, a LCD display screen and a keypad with 6 keys (up, down, right, left and ESC and OK keys). The display screen and keypad presented with the PLC are the viewing elements for states, defaults and other messages. They also are used to programmed the various data values. In standard operational mode, date and time, program version or the number of cycles are displayed.

Keys ▲ ▼ are used to change screen vertically, move within the program, as well as increase or decrease the value of a data.

Keys ◄ ► are used to change screen horizontally, move onto the various characters displayed for a programmed value (tens, units, seconds…).

The ESC key is to be used to go back to the previous menu level in case a mistake in entering occurred at one stage.

The OK key is used to call for a programmed value, to validate it, or enter the selected menu.

Constants and connectors (Co)

Constants and connectors (= Co) represent inputs, outputs, flags and constant voltage levels

Digital inputs

Digital inputs are identified with an I. The number of the digital inputs (I1, I2, ...) corresponds with the number of the input connectors of the LOGO! Basic and of the connected digital modules, in the order of their installation

Analog inputs

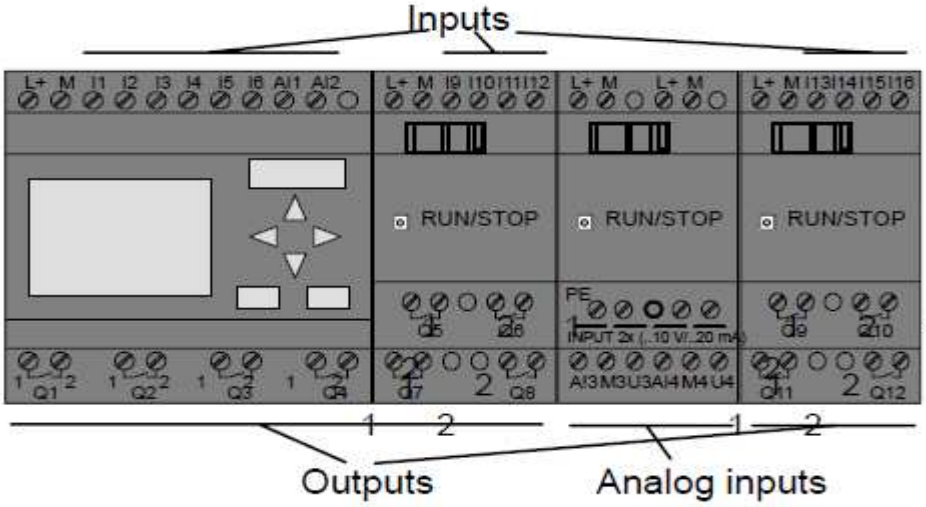
We can used I7 and I8 inputs to analog signal, which can also be programmed for use as AI1 … AI8 inputs, analog flags AM1….AM6, and analog output AQ1 and AQ2.

Digital outputs

Digital outputs are identified by the character Q. The output numbers (Q1, Q2, ... Q16) correspond with the numbers of the output connectors at the LOGO!

Analog outputs

Analog outputs are identified by the letters AQ. Two analog are available, namely AQ1 and AQ2.

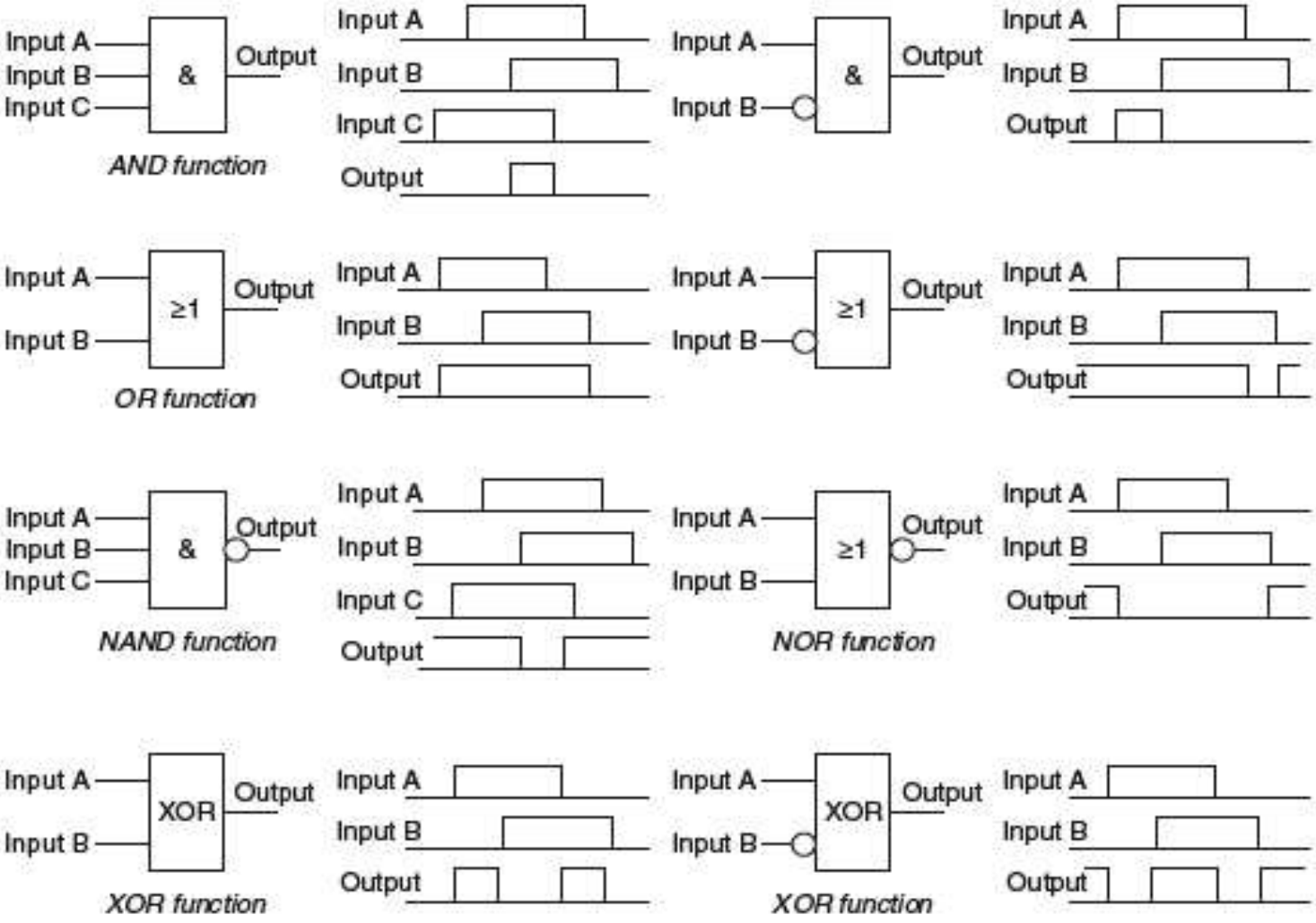


Flag blocks

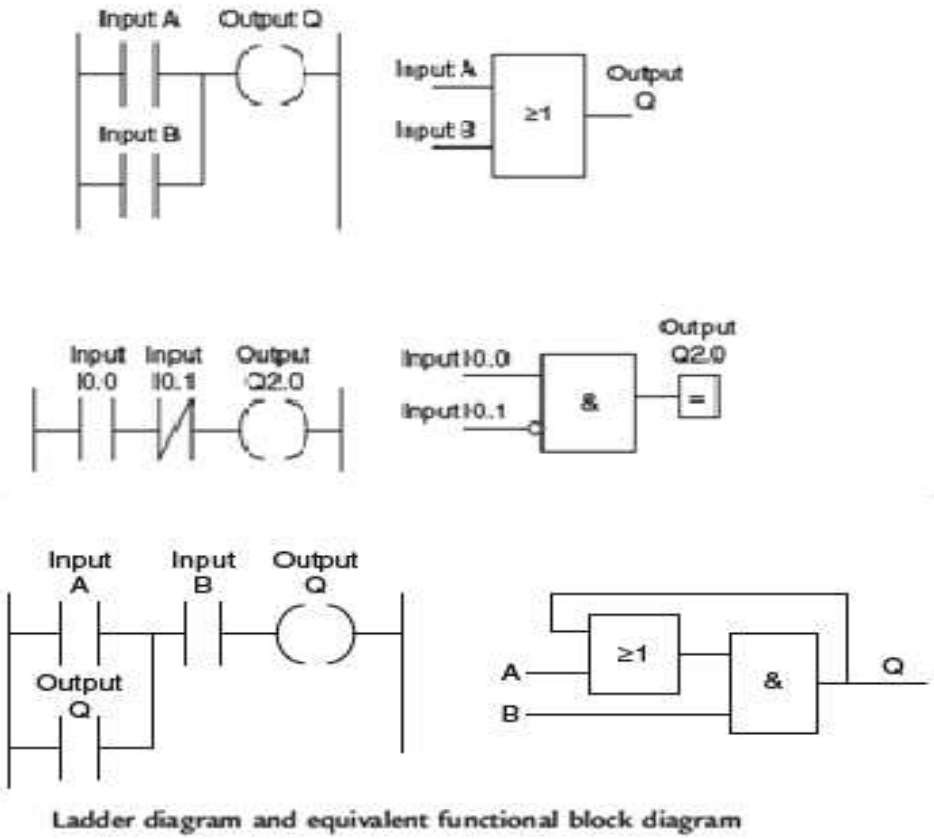
Flag blocks are identified by the letters M or AM . These are virtual outputs, which output the value of their inputs. LOGO! provides 24 digital flags M1 ... M24 and 6 analog flags AM1 ... AM6.

General Function (GF)

Basic functions represent simple logical elements of Boolean algebra. The GF list contains the basic function blocks you can use for your circuit program. The following basic functions are available:



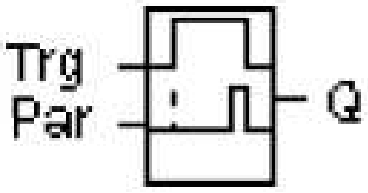
Ladder Diagram and equivalent functional block diagram



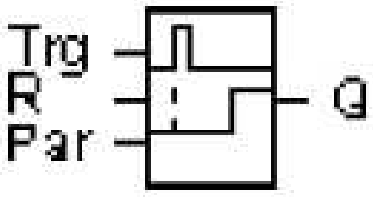
Ladder diagram and equivalent functional block diagram

Special functions list – SF

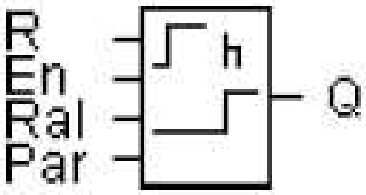
When you create your circuit program in LOGO!, you find the special function blocks in the SF list. The following some of SFs



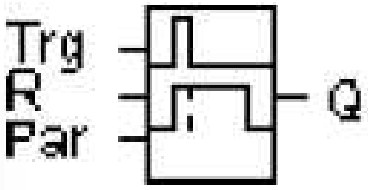
On-delay



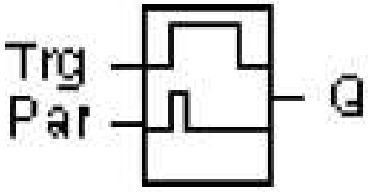
Retentive on-delay



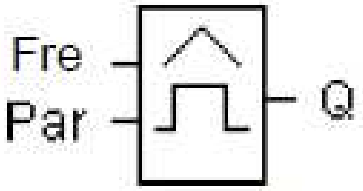
Hours counter



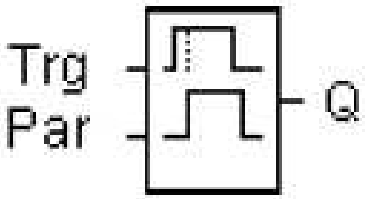
Off-delay



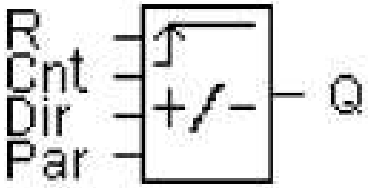
Wiping relay (pulse output)



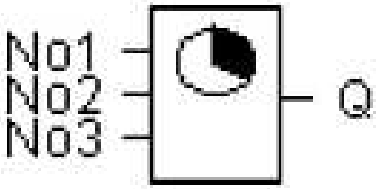
Threshold trigger



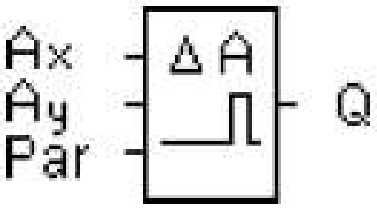
On-/Off-delay



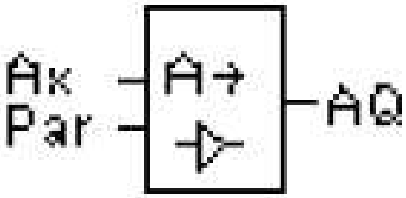
Up/down counter



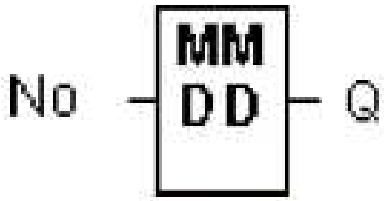
Weekly timer



Analog comparator



Analog amplifier



Yearly timer

التجربة العاشرة

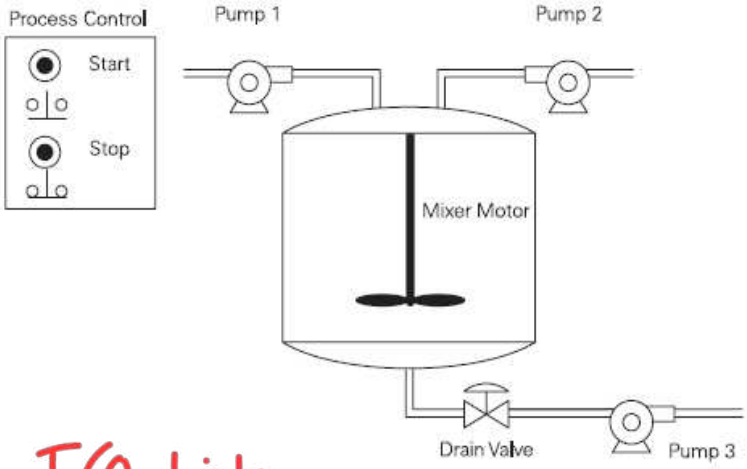
LOGO LOGIC Example

Problem 1

A tank will be filled with two chemicals, mixed, and then drained. When the Start Button is pressed, the program starts pump 1 runs for 5 seconds, filling the tank with the first chemical, then shuts off. The program then starts pump 2, runs for 3 seconds filling the tank with the second chemical. After 3 seconds pump 2 shuts off. The program starts the mixer motor, and mixes the two chemicals for 60 seconds. The program then opens the drain valve, and starts pump 3.

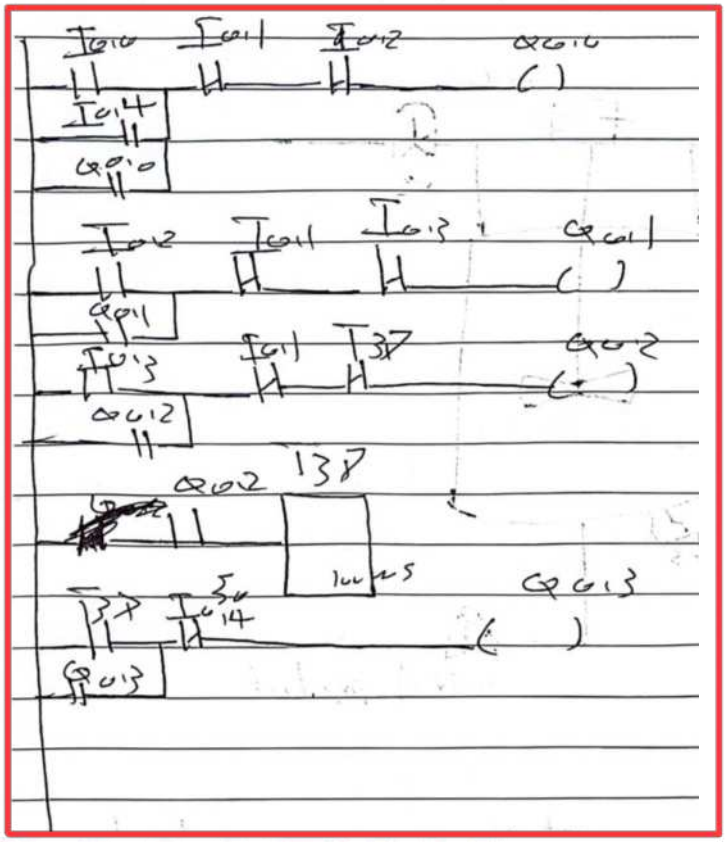
Pump 3 shuts off after 8 seconds and the process stops. A manual Stop switch is also provided to stop process at any time .

- Write a ladder diagram to control the system
- Write program in Functional Block
- Input / output table

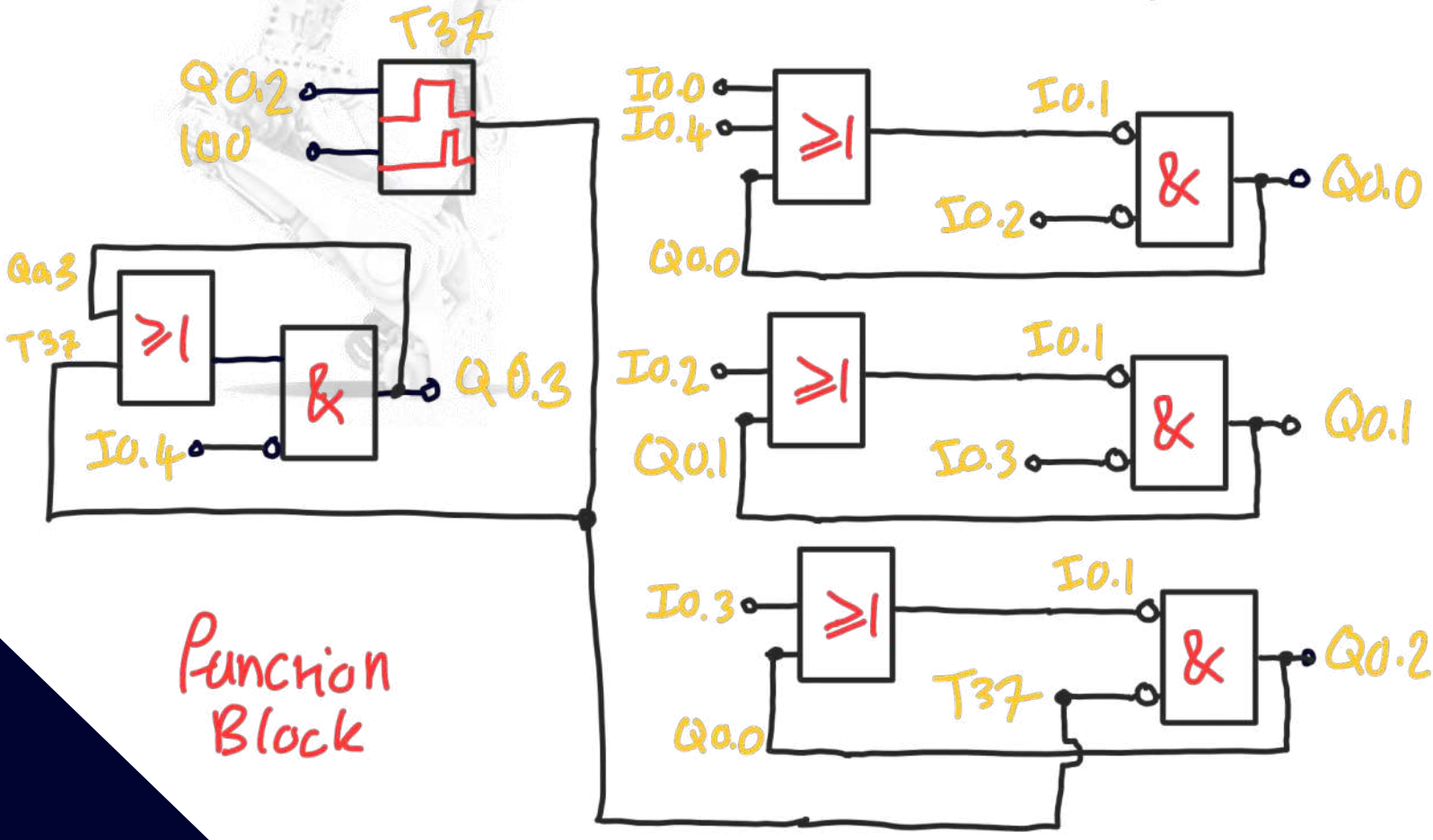


I/O list

Input		output	
I0.0	start	Q0.0	V1
I0.1	stop	Q0.1	V2
I0.2	level A	Q0.3	Motor
I0.3	level B	Q0.4	V3
I0.4	LLS		mixed product



ladder diggram

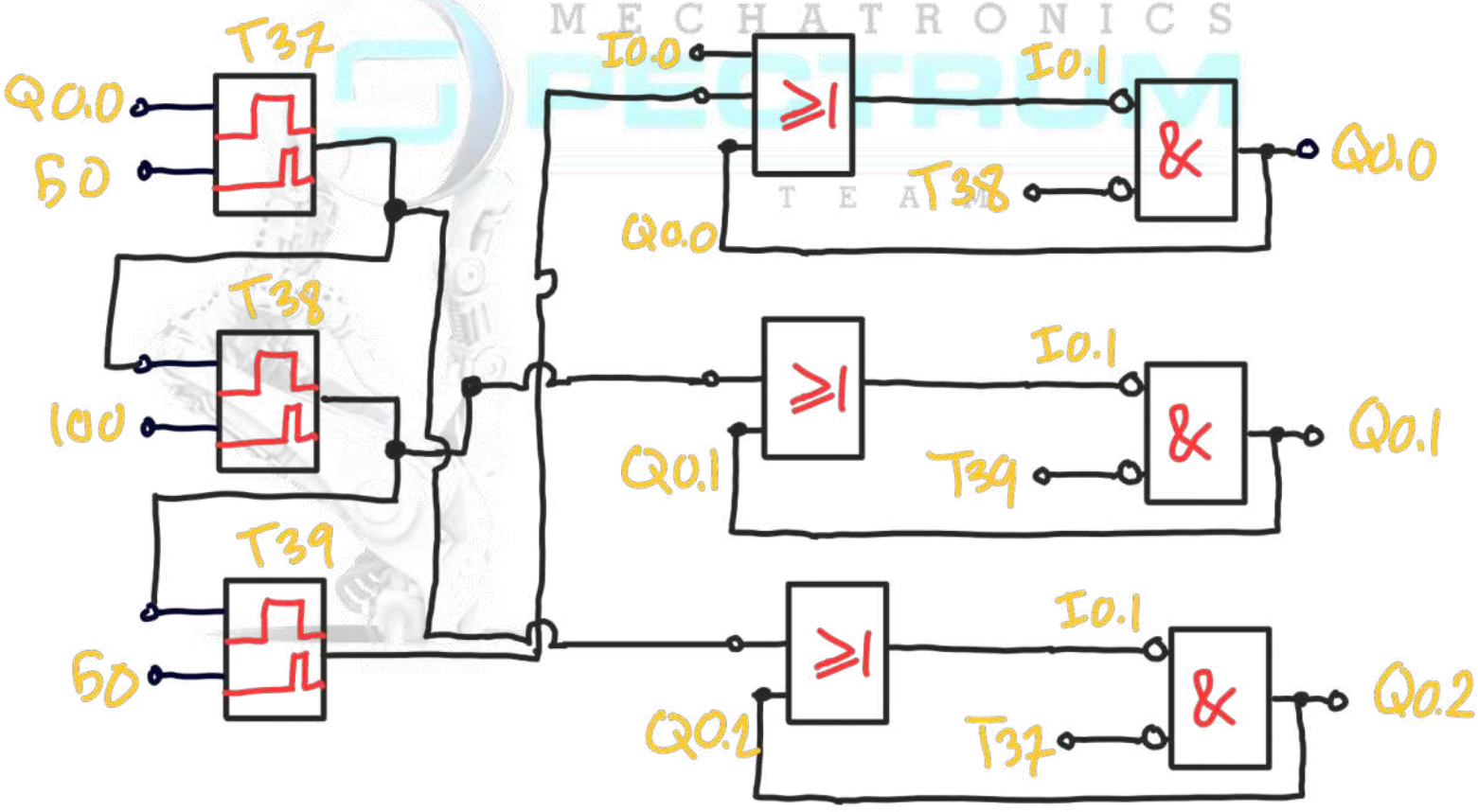


Function Block

Problem 2

Control a traffic light system using plc with the following sequence:

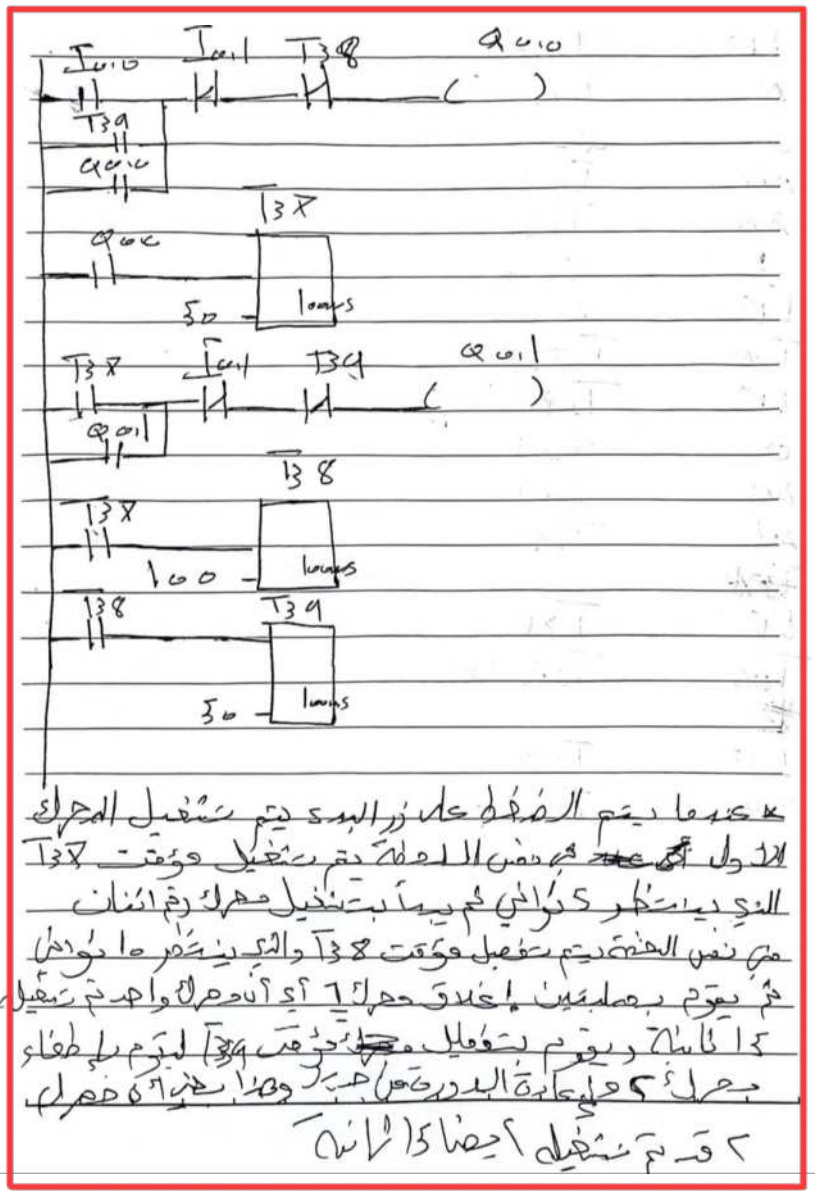
1. The red light is on for a 5 seconds
2. The red light turns off and the yellow light is turned on for a 2 seconds
3. The yellow light turns off and the green light is turned on for a 4 seconds
4. The red light turns on again



Function Block

```
LD      I0.0
AN      T39
O       Q0.0
AN      I0.1
AN      T38
=       Q0.0
LD      Q0.0
TONR    T37, 50
LD      T37
O       Q0.1
AN      I0.1
AN      T39
LD      T37
TON     T38, 100
LD      T38
TON     T39, 50
```

instruction list



ladder

عندما يتم الضغط على زر البدء يتم تشغيل المحرك الأول حيث في نفس اللحظة يتم تشغيل مؤقت T37 الذي يبدأ العد 5 ثوانٍ ثم يبدأ بتشغيل محرك رقم اثنان في نفس اللحظة يتم تشغيل مؤقت T38 والذي يبدأ العد 10 ثوانٍ ثم يقوم بعمليتين إغلاق محرك 1 أي أن المحرك واحد تم تشغيل 5 ثانية ويقوم بتشغيل مؤقت T39 ليقوم بإطفاء محرك 2 وإعادة البدء من جديد وهذا يعني أن محرك 2 قد تم تشغيل أيضا 5 ثانية

بسواعدنا نبنيها
MECHATRONICS SPECTRUM TEAM

المصادر والمراجع

التجربة الاولى

https://www.youtube.com/watch?v=Uo5GC2AQYSk&list=PLxYPveP_EstGIbTPfVoYgNaZEJTZlaw23

التجربة الثانية

https://instrumentationtools.com/tag/siemens-s7-200-plc-programming-examples-pdf/

التجربة الثالثة

https://instrumentationtools.com/timers-in-plc/

التجربة الرابعة

https://instrumentationtools.com/plc-program-for-iec-timers-ton-tof-tp-tonr-used-in-s7-1200/

التجربة الخامسة

https://forumautomation.com/t/how-to-connect-and-download-a-program-in-siemens-s7-200-plc/8984

التجربة السادسة

https://youtu.be/EzucwY80r44?si=wF0eCDyhJw-lBxPc

التجربة السابعة

https://youtu.be/gT7ESD824z8?si=oRvWFpGyqnh_mVZh

التجربة الثامنة

https://www.pdfsupply.com/blog/index.php/2022/05/23/examples-of-plc-inputs-and-outputs/

التجربة التاسعة

1.

https://youtu.be/_lgkc3ITRRI?si=gxhu5oaMkuIDgLqq

2.

https://www.siemens.com/global/en/products/automation/systems/industrial/plc/logo/logo-basic-modules.html

في الختام اسأل الله ان يوفقنا وإياكم ويهدينا سبيل الرشاد



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